

XQuality project

Comparison between Normal production & Ab-normal production

Case of the changes in the material for grade 316L instead of the 303 stainless steel reference grade



Raw Data

Raw Data



Batch #1 & #2 :

- **Part #7 to Part #60** : Normal process conditions
- Part #61 to Part #70 : Feedrate reduced to 80%
- **Part #71 to Part #75** : 316L grade instead of 303 grade
- Part #76 to Part #77 : Normal process conditions for tool wear changes verification (if any)
- Part #78 to Part #81 : Tool references mistakes (voluntary)

Machining operations numbering



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Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

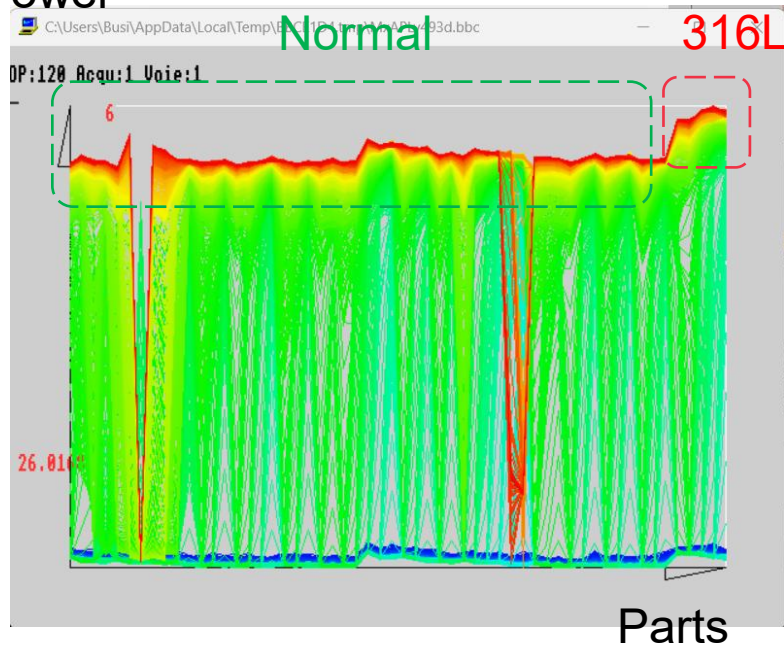
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Intermediate report Project n°316285 - Date 26/01/26 Page 91/188
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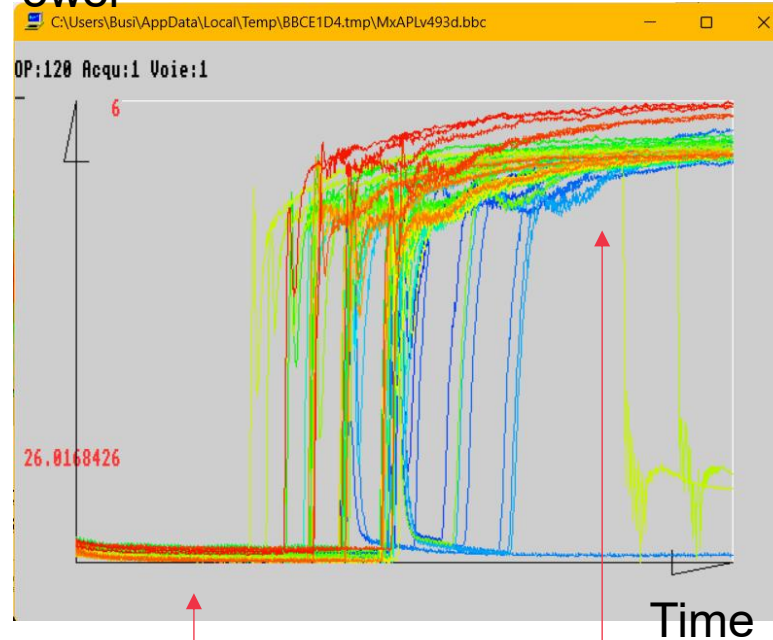
Machining OP 120

Spindle Power : Ai1

Power



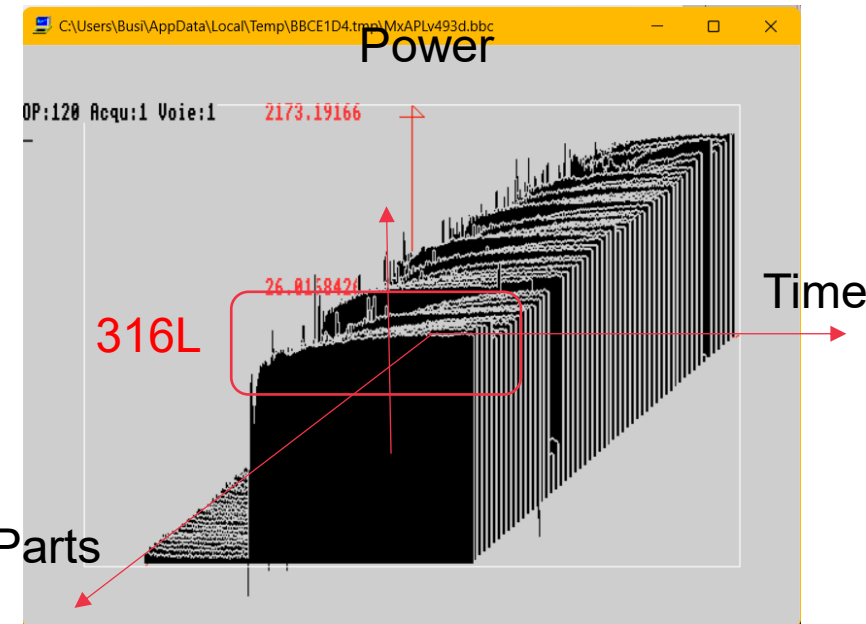
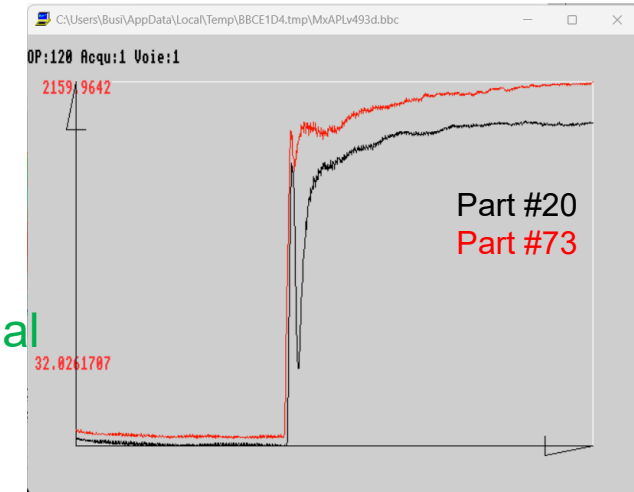
Power



Before machining

In machining

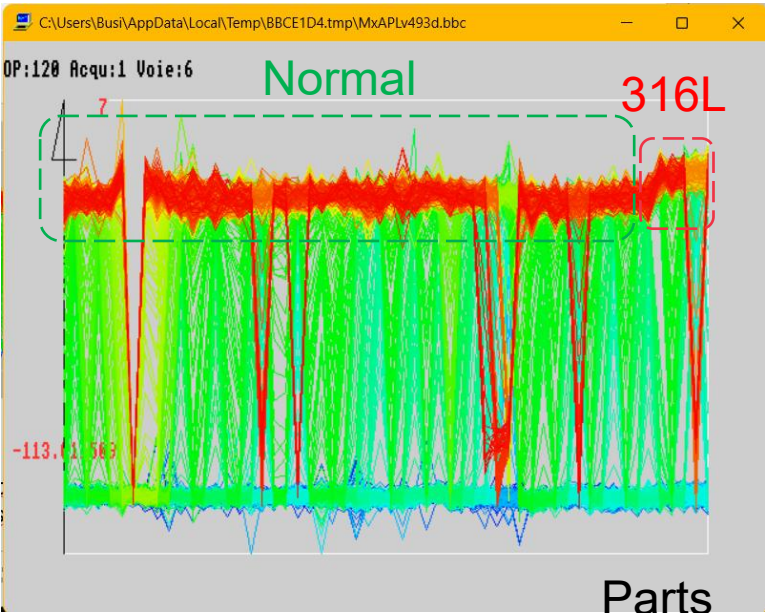
Normal
316L



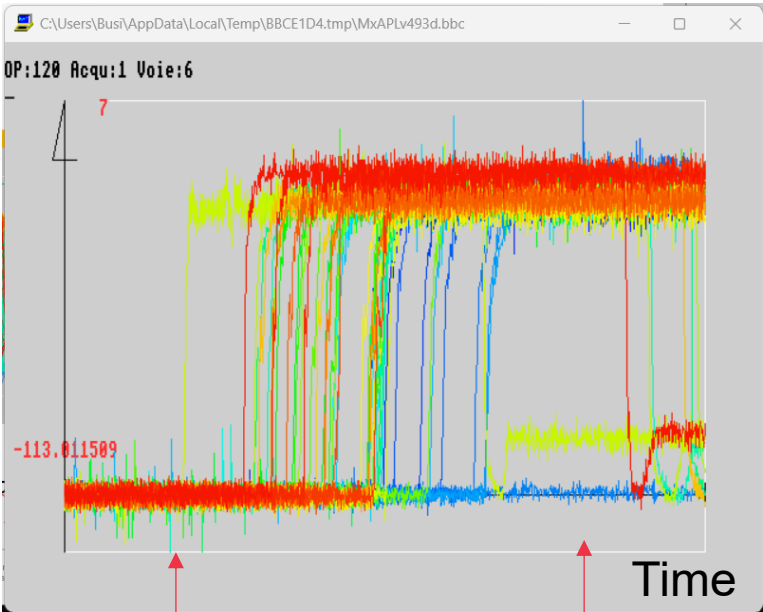
Rem : Temporal shifts sometimes !
Spindle power increases when 316L grade => OK

Cutting force : Fz1

Force



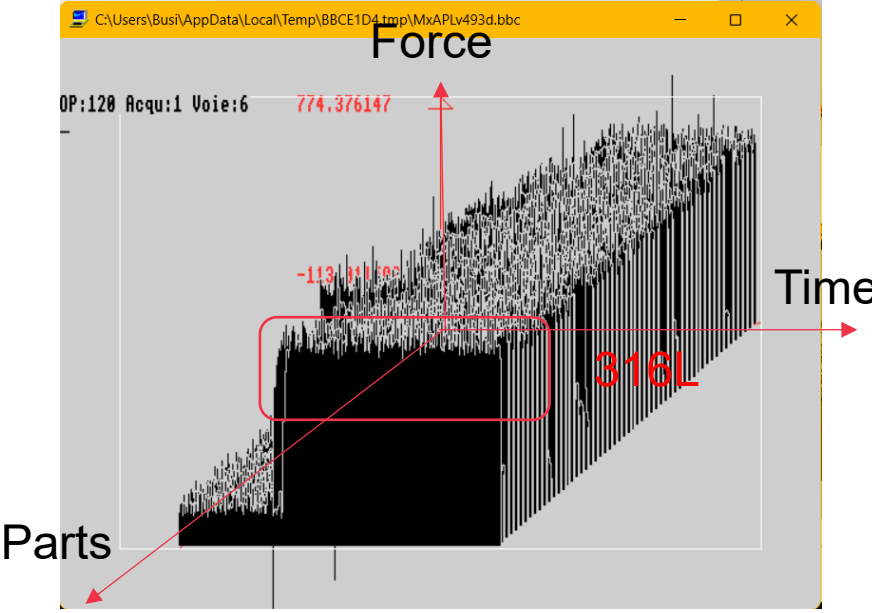
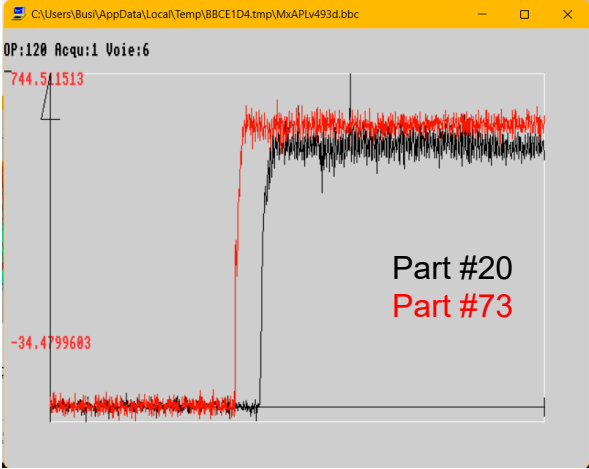
Force



Before machining

In machining

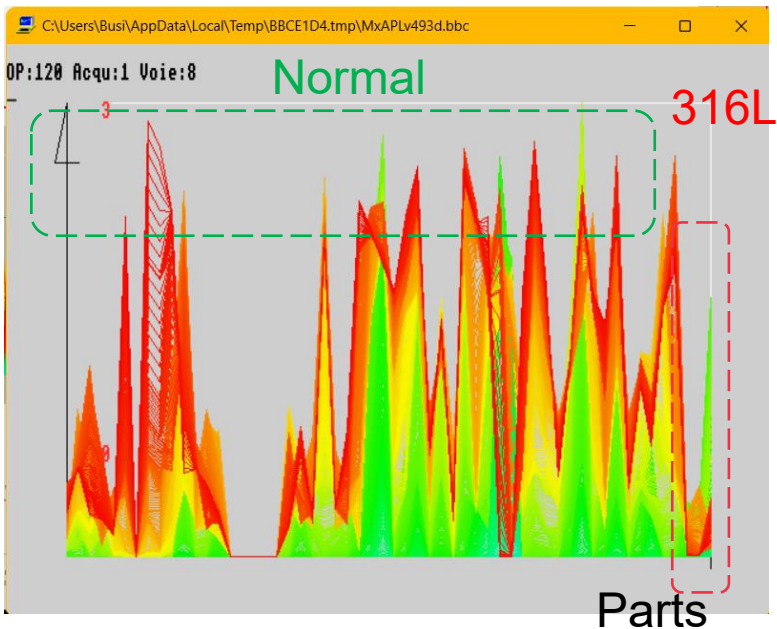
Normal
316L



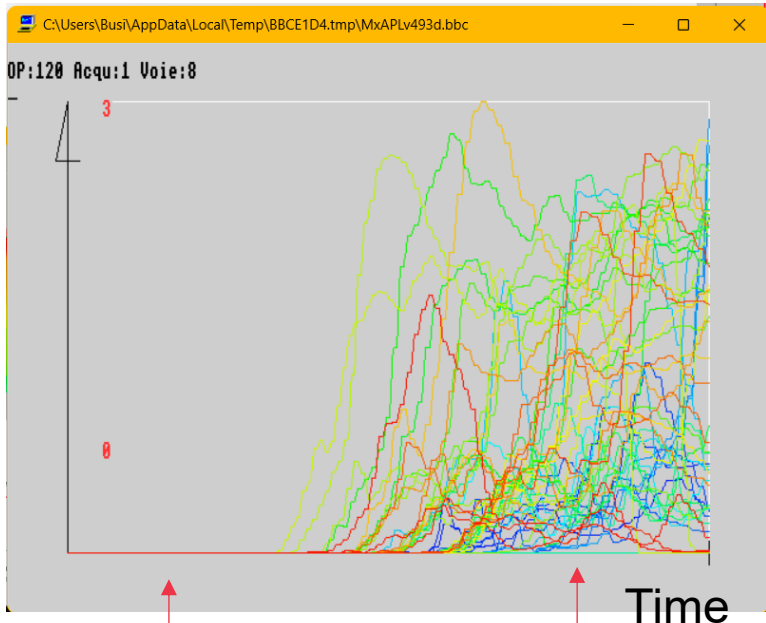
Rem : Temporal shifts sometimes !
Cutting Force increases when 316L grade => OK

Vib : Vib1

RMS



RMS

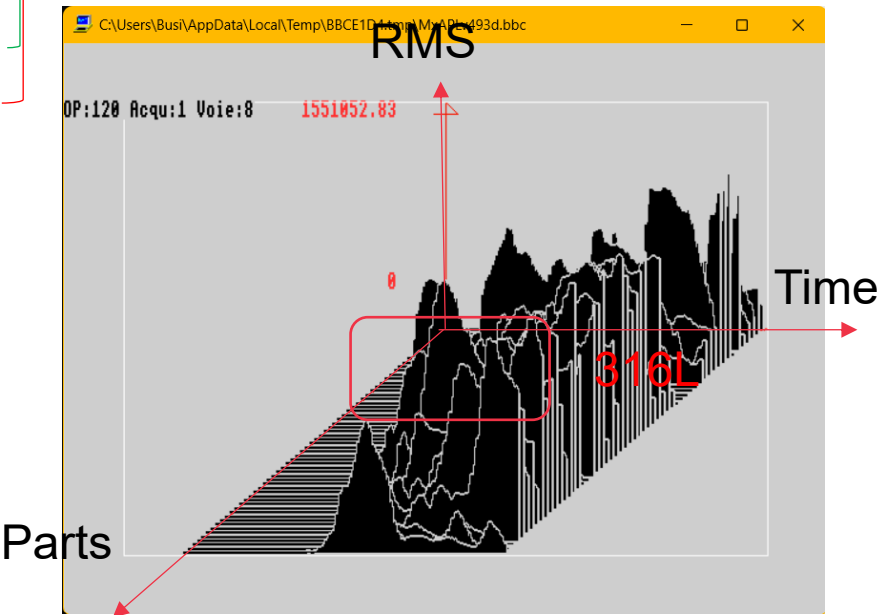
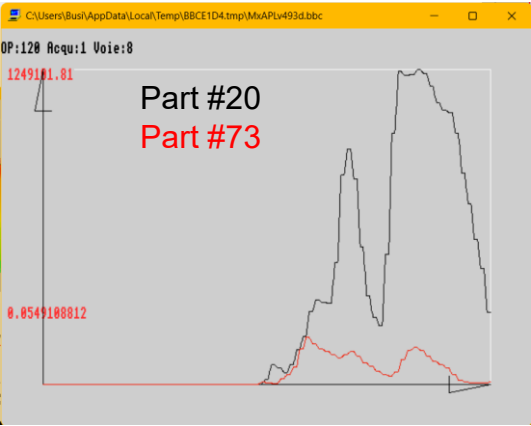


Before machining

In machining

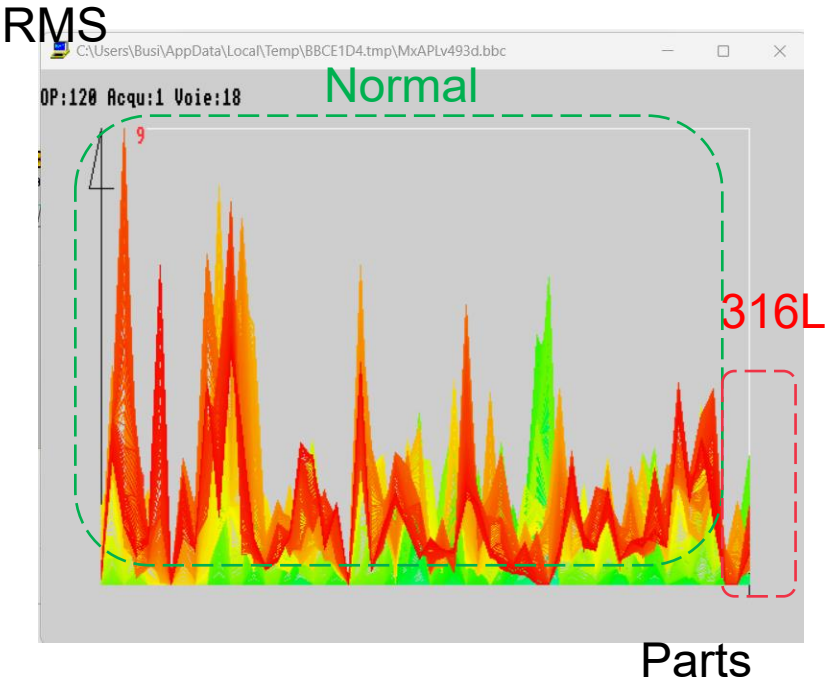
Normal

316L

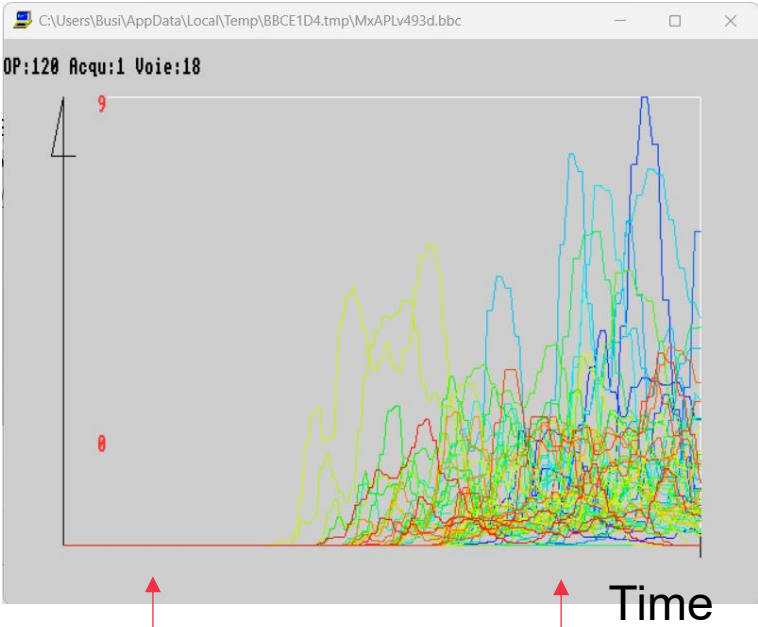


RMS decreases when 316L grade

Audio : Vib 11

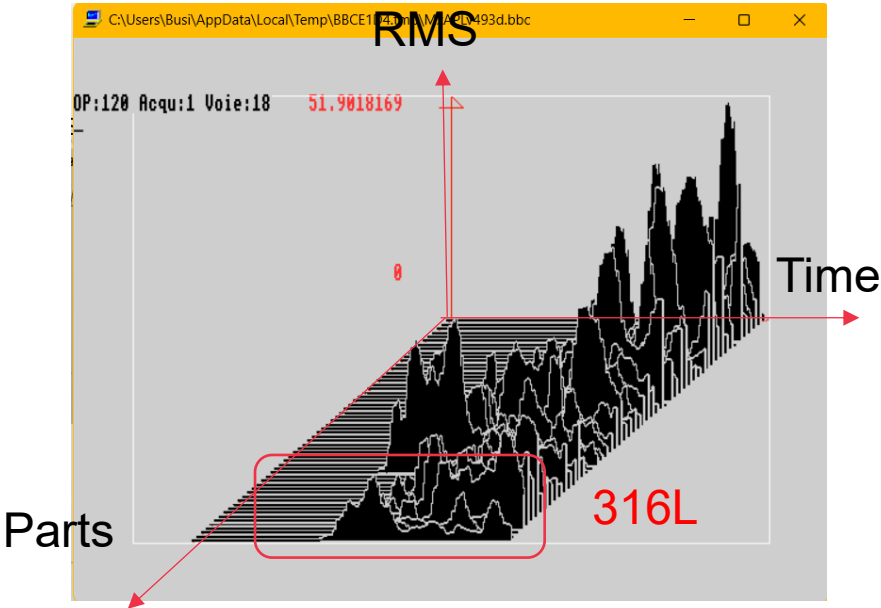
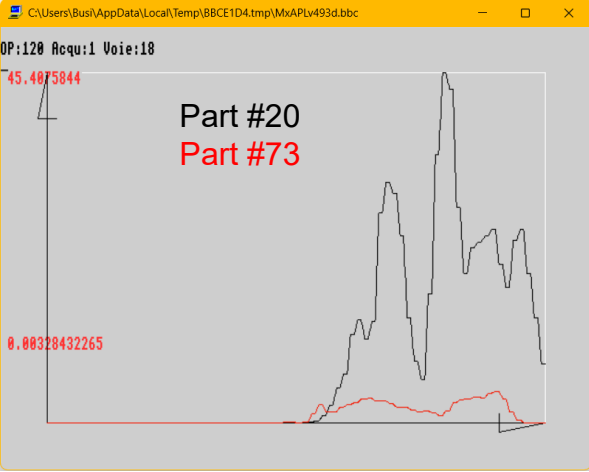


RMS



Before machining

In machining



RMS decreases when 316L grade

Sensors number (for my viewing software) :

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- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

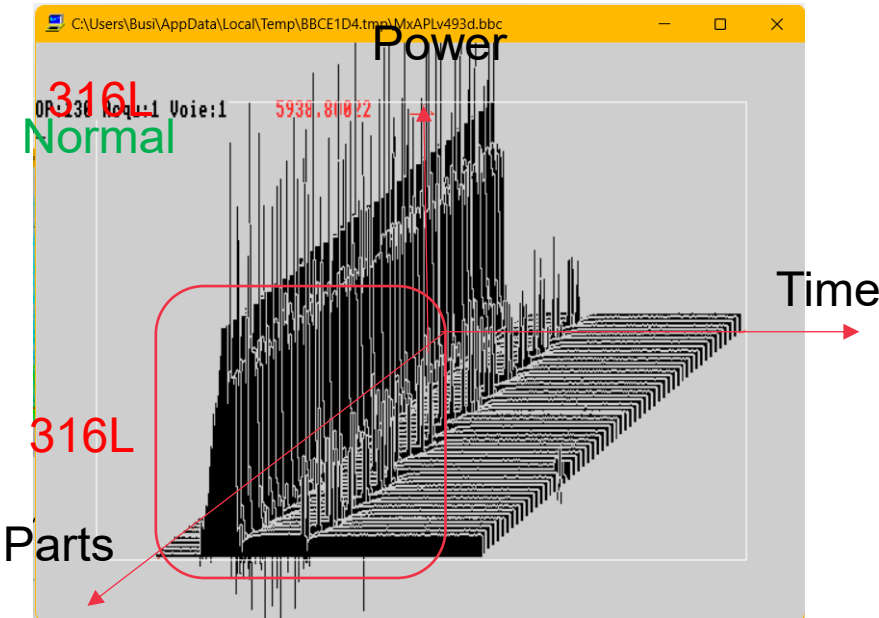
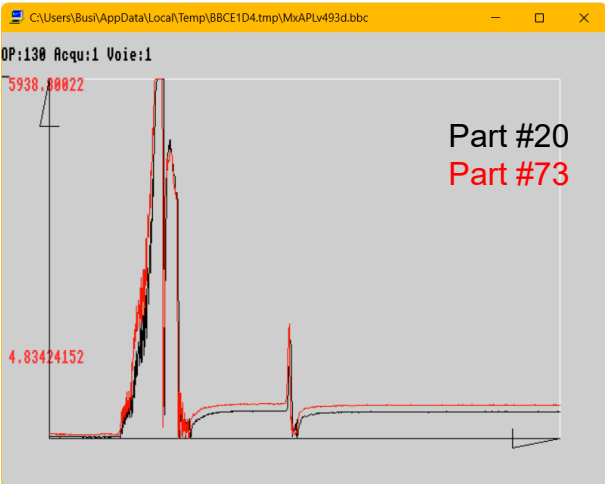
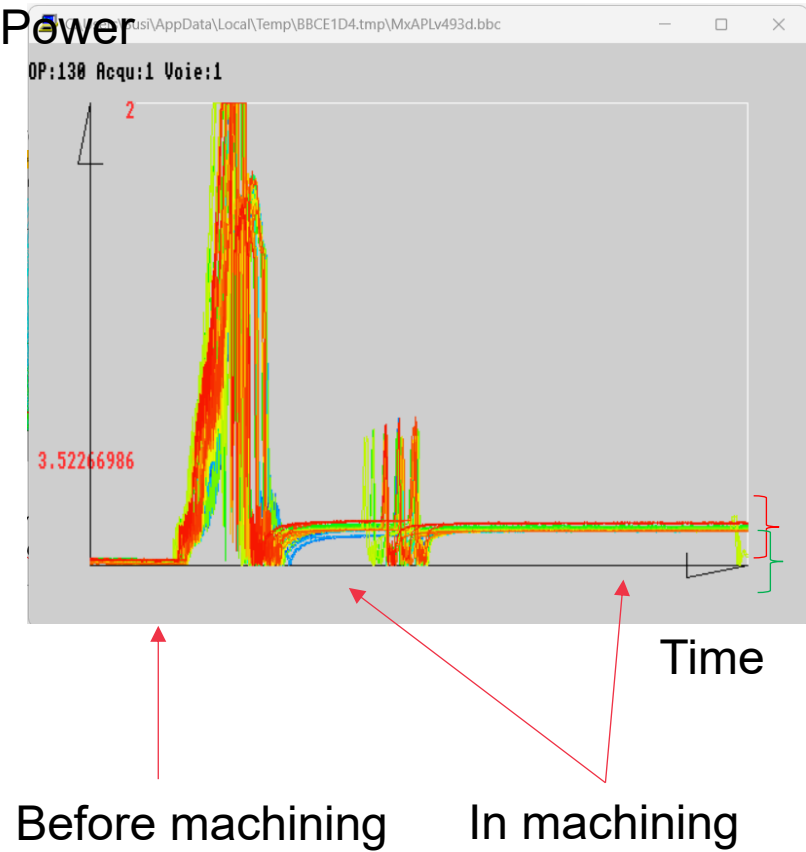
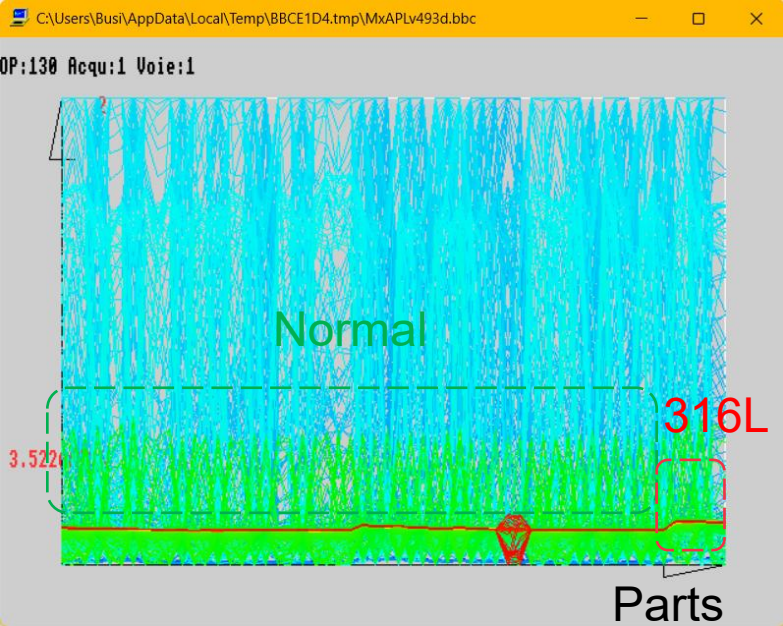
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Intermediate report Project n°316285 - Date 26/01/26 Page 91/188
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Machining OP 130

Spindle Power : Ai1

Power

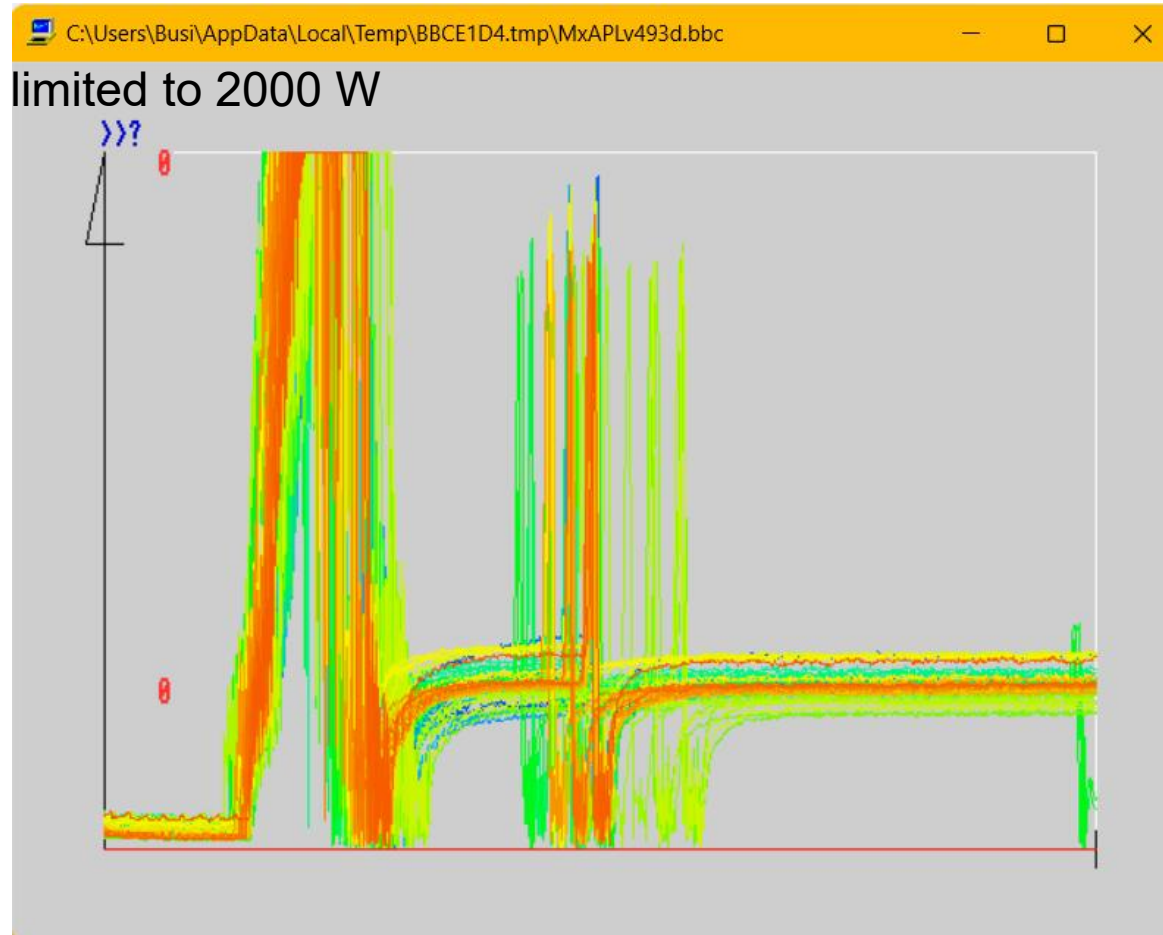


Power increases when 316L grade => OK

Spindle Power : Ai1



Power limited to 2000 W



Rem : Temporal shifts sometimes !

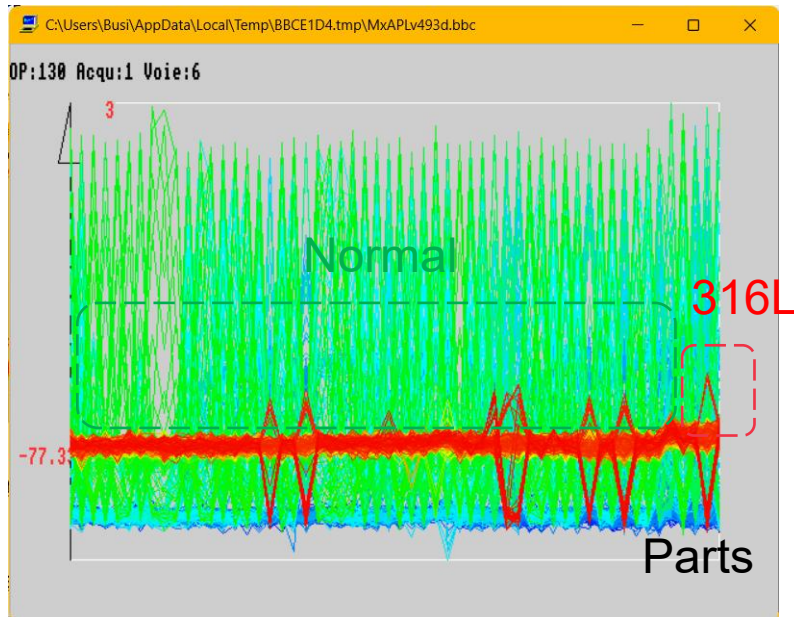
With a vertical zoom => Small increase of spindle Power when 316L grade => OK

Cutting force : Fz1



Part #20
Part #65

Force



Force

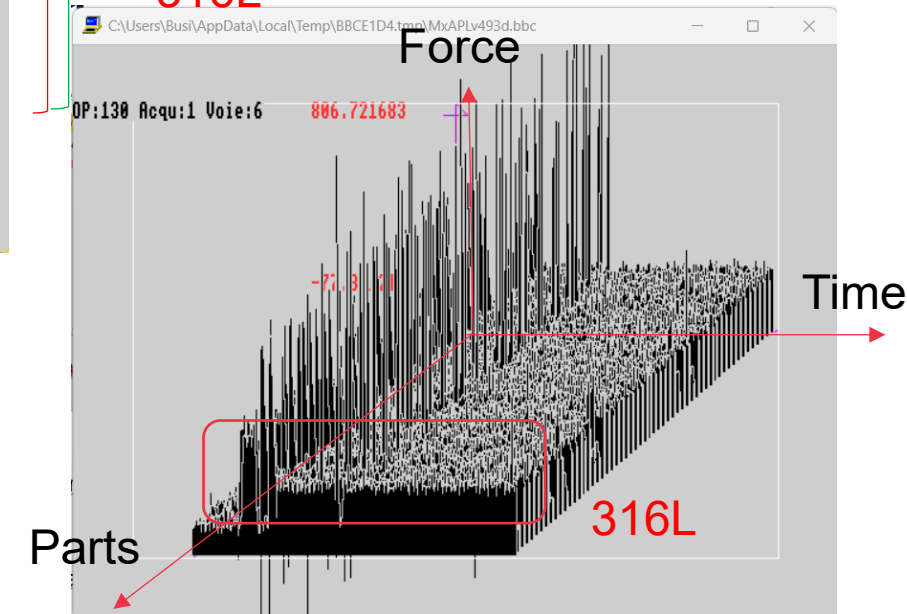


Before machining

In machining

Normal

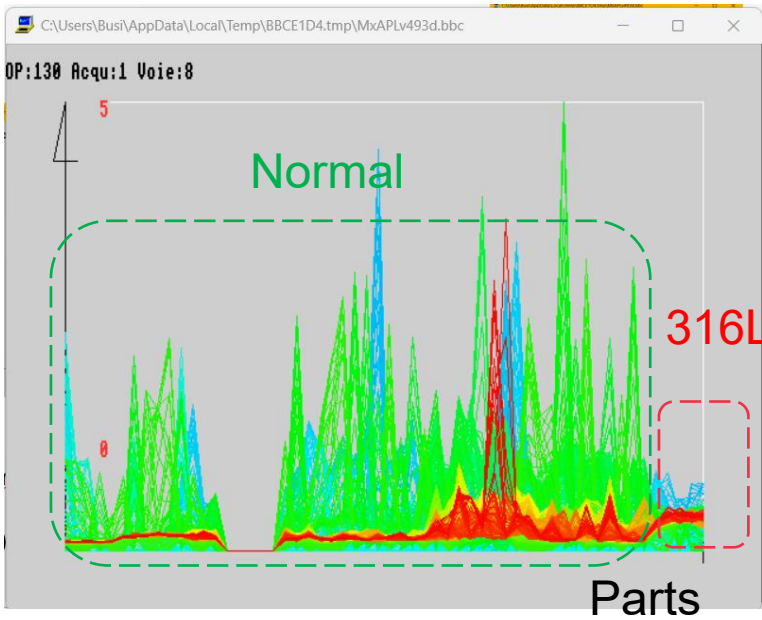
316L



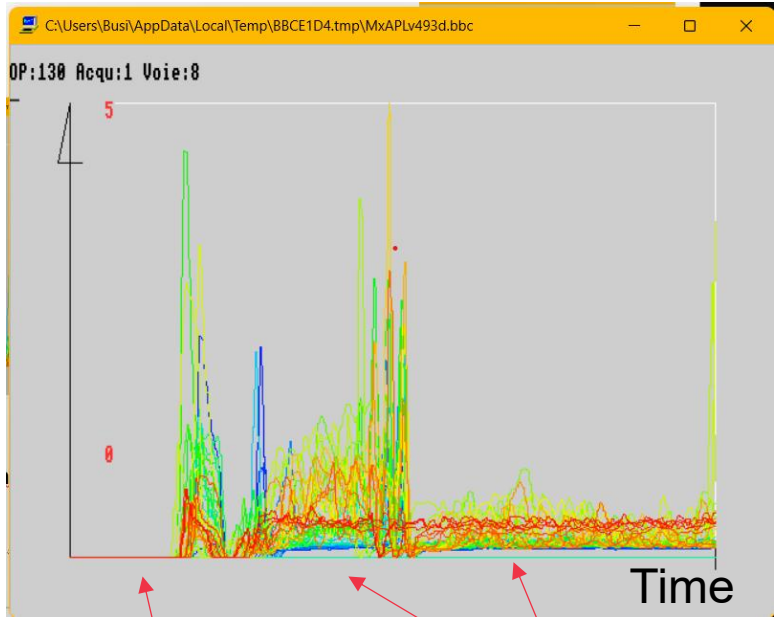
*Rem : Temporal shifts sometimes !
Cutting Force increases a little when 316L grade => OK*

Vib : Vib1

RMS



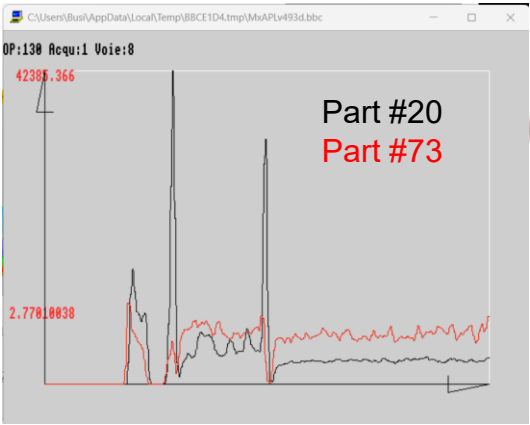
RMS



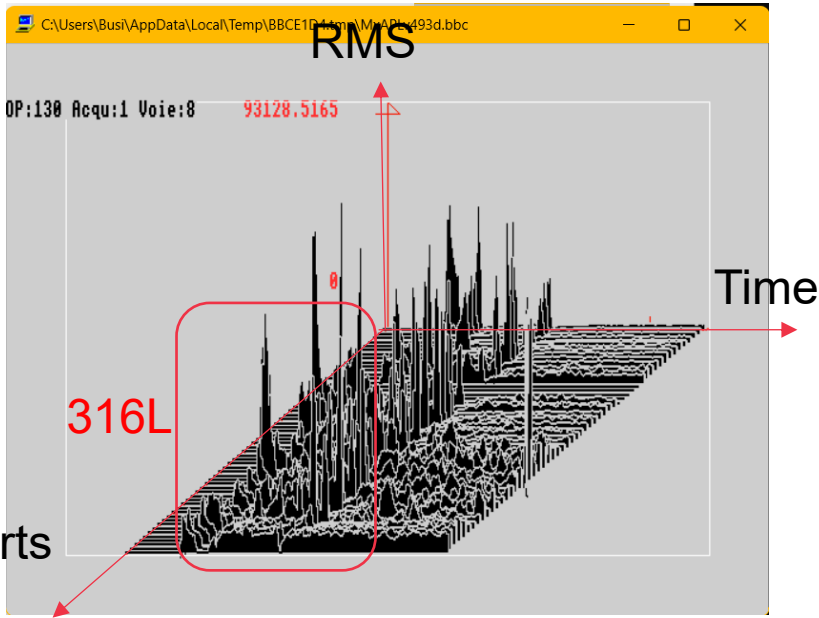
Before machining

In machining

316L



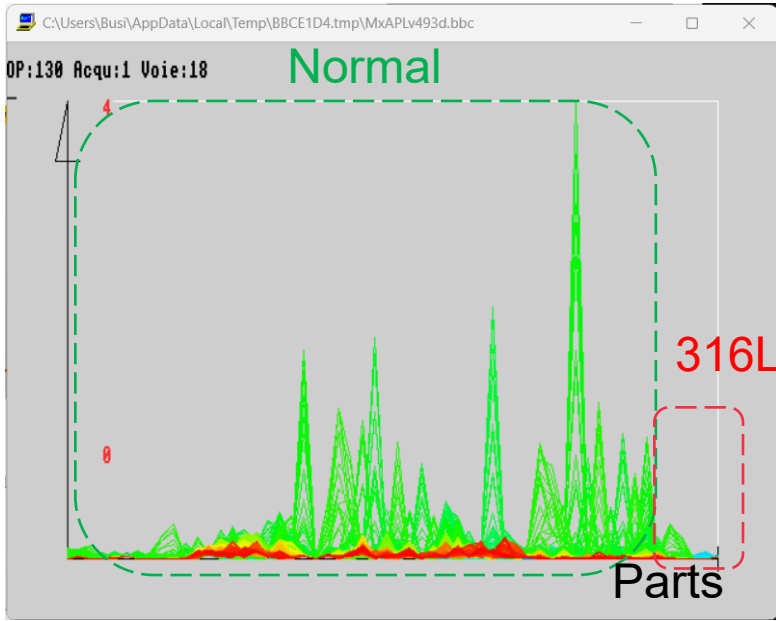
Normal



RMS increases when 316L grade

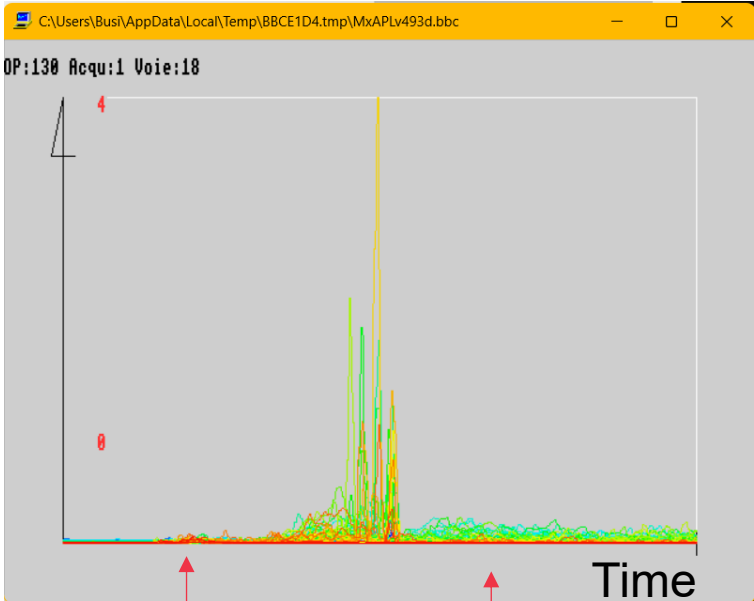
Audio : Vib 11

RMS



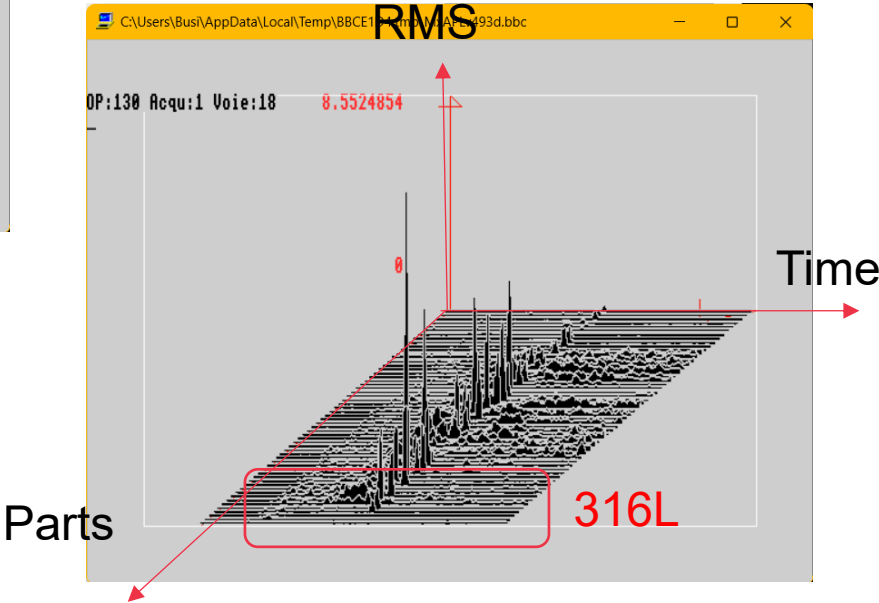
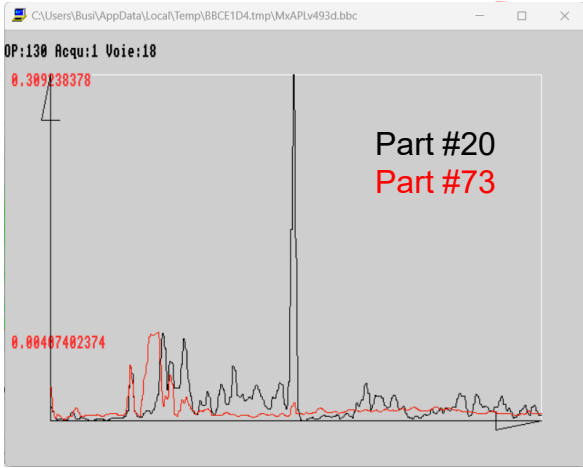
RMS audio decreases when 316L grade => OK

RMS



Before machining

In machining



Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

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#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

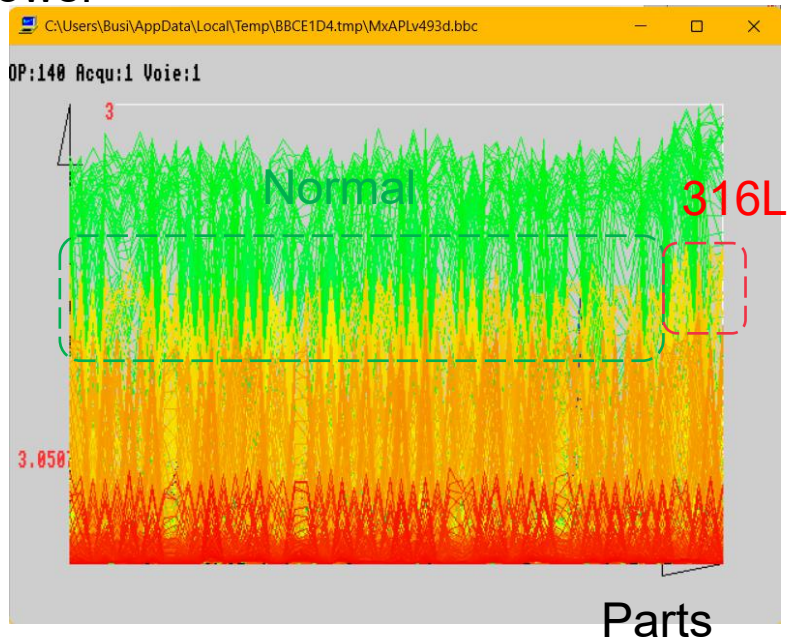
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Intermediate report Project n°316285 - Date 26/01/26 Page 91/188
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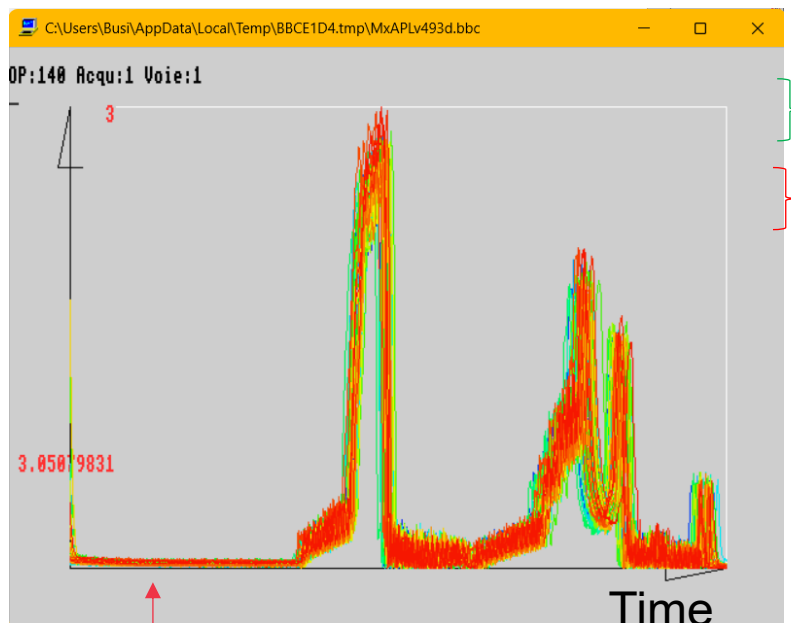
Machining OP 140

Spindle Power : Ai1

Power



Power

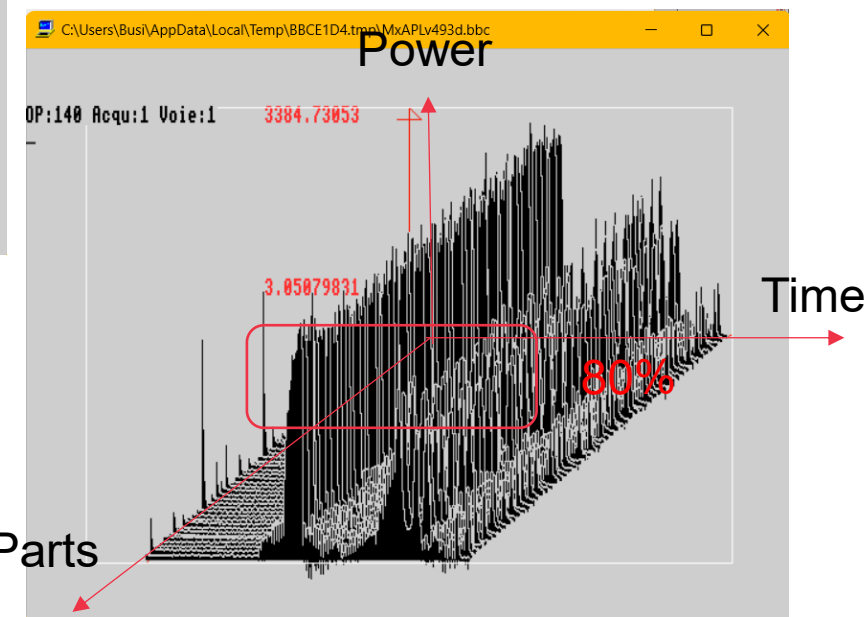
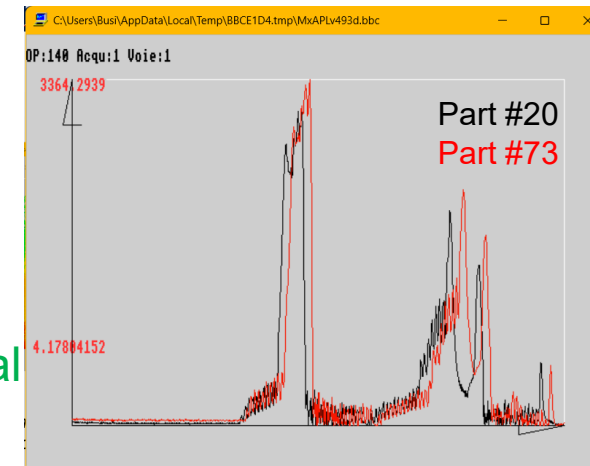


Before machining

In machining

Normal

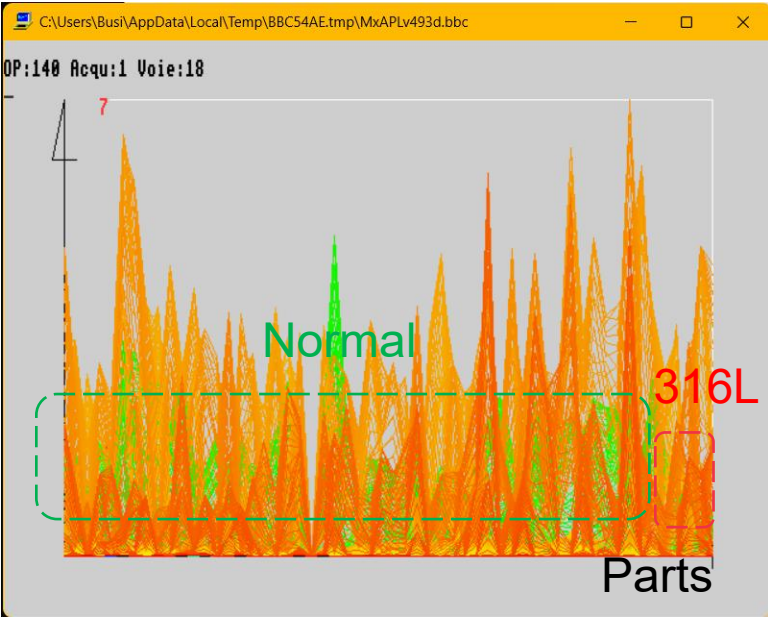
80%



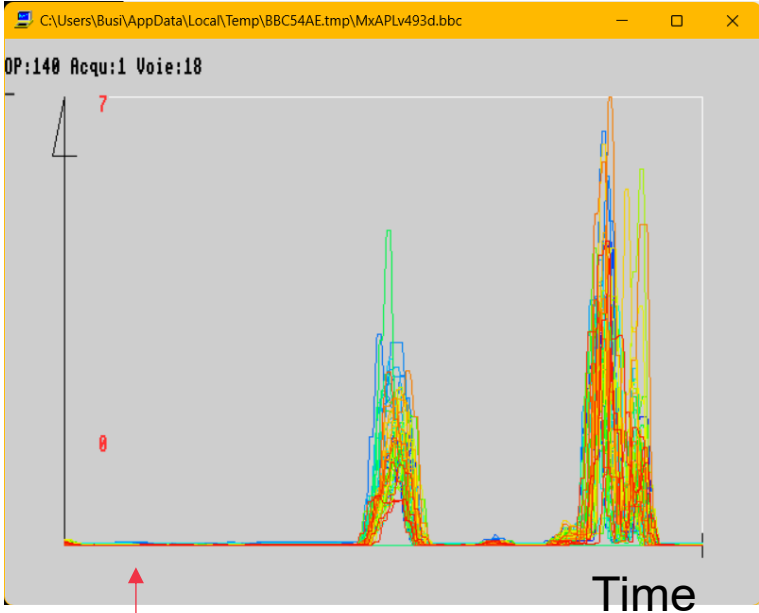
Spindle power increases a little when 316L grade => OK

Audio : Vib 11

RMS

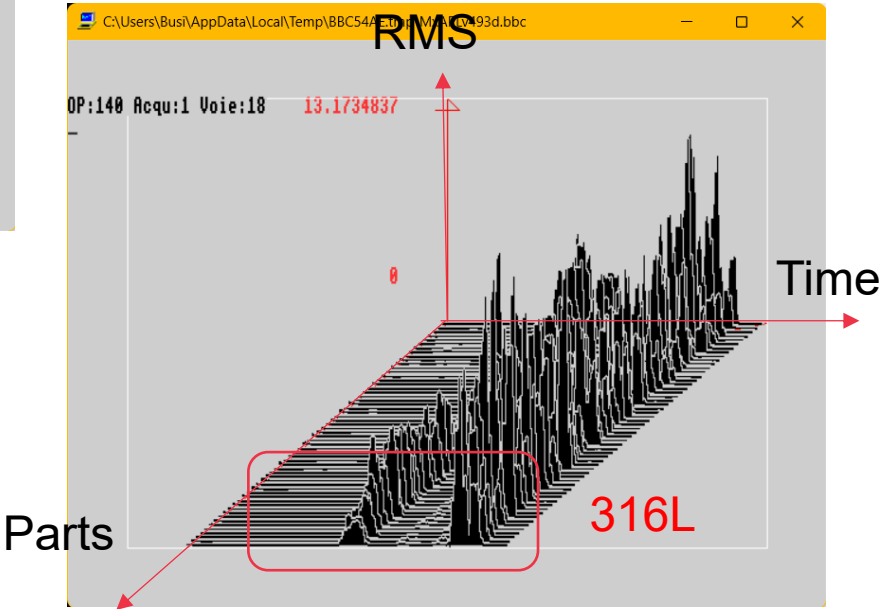
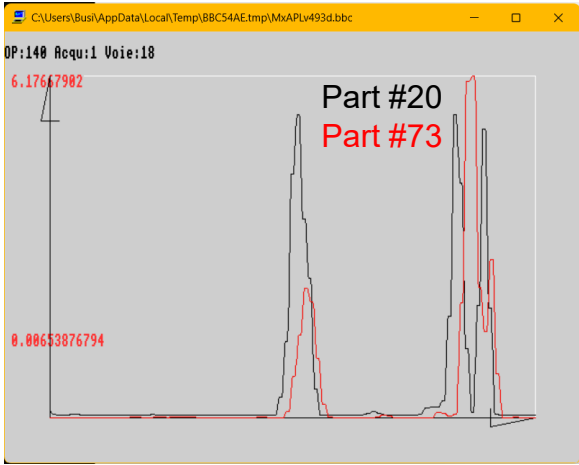


RMS



Before machining

In machining



RMS seems to be lower on the 1st peak, and the same on the second peak

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

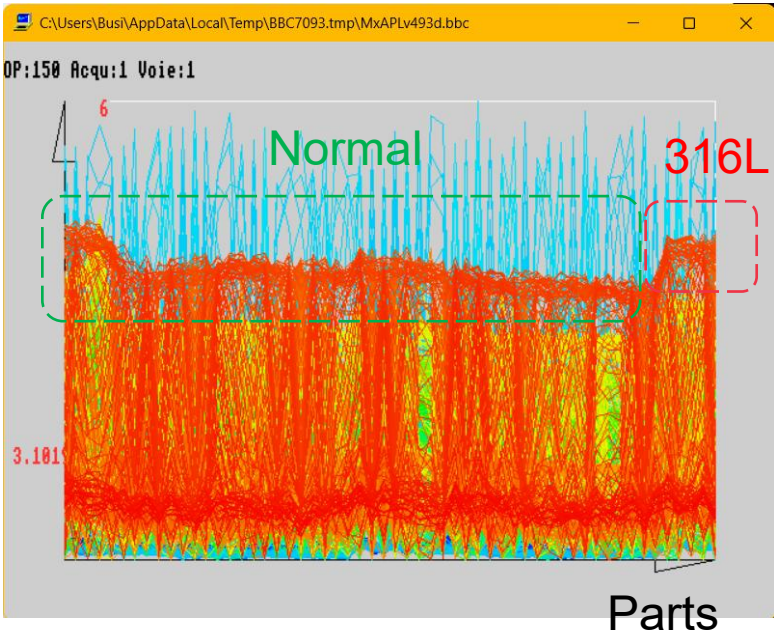
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<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
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<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

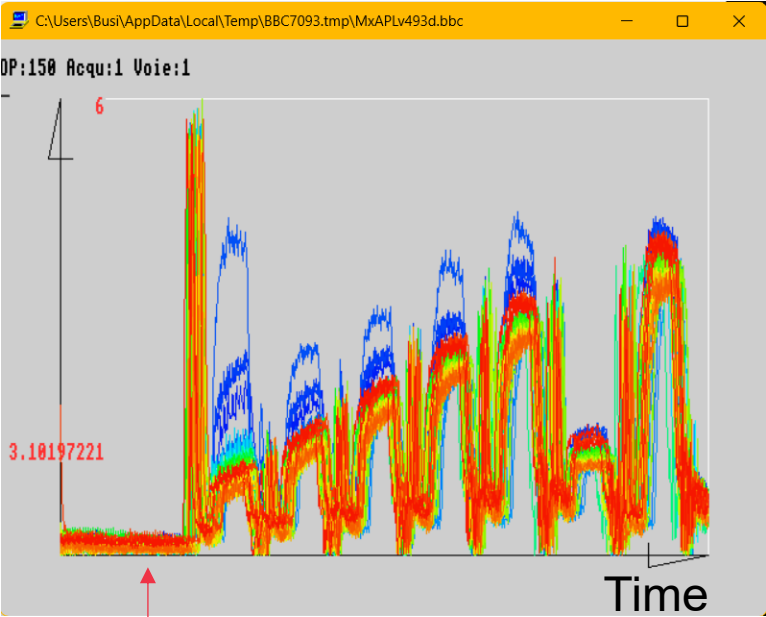
Machining OP 150

Spindle Power : Ai1

Power

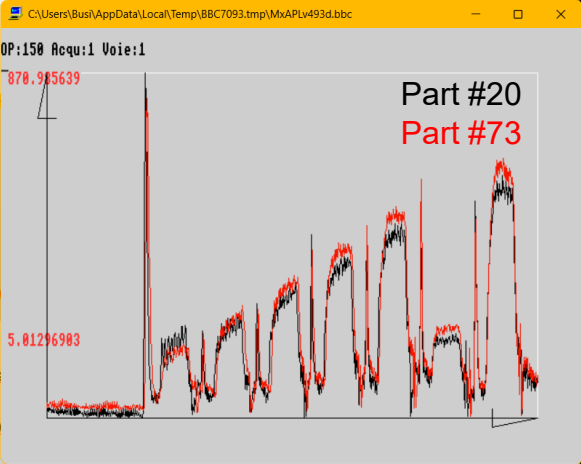


Power

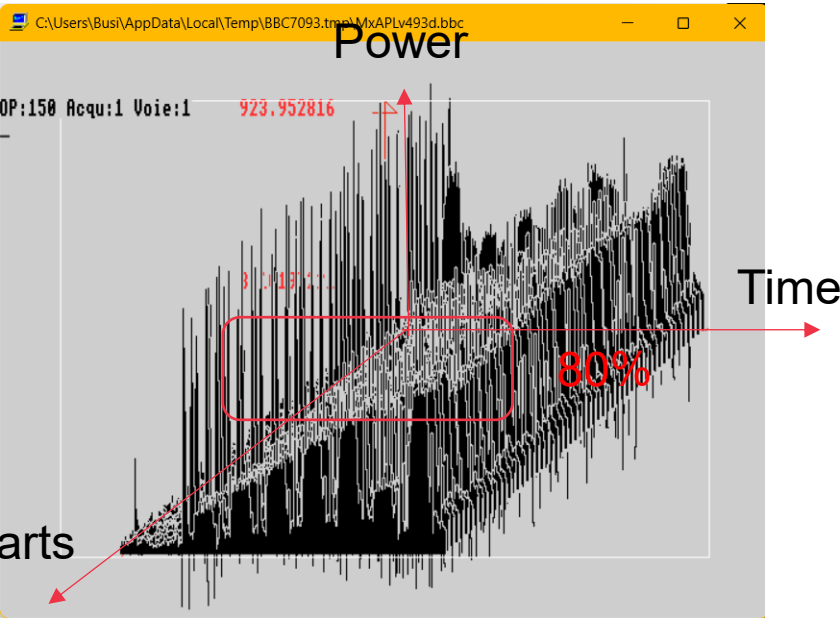


Before machining

In machining



316L
Normal



Spindle power increases when 316L grade => OK

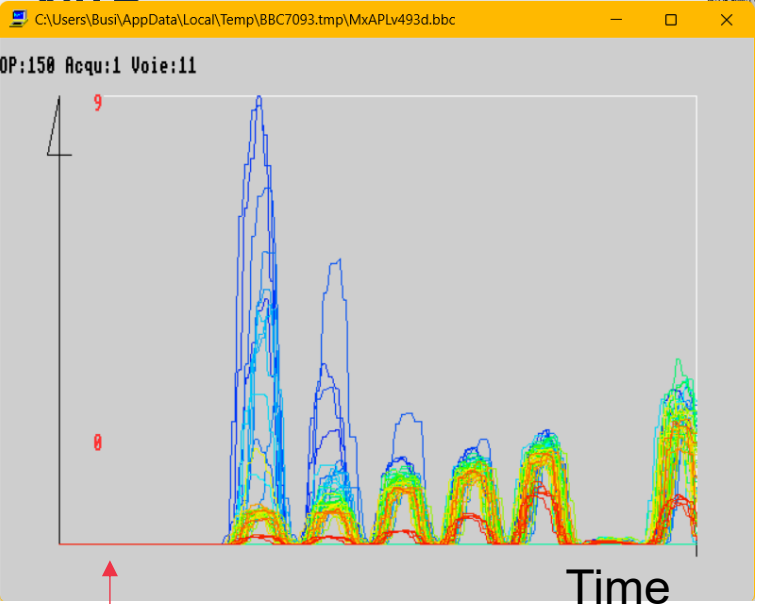
Vib : Vib 4

RMS



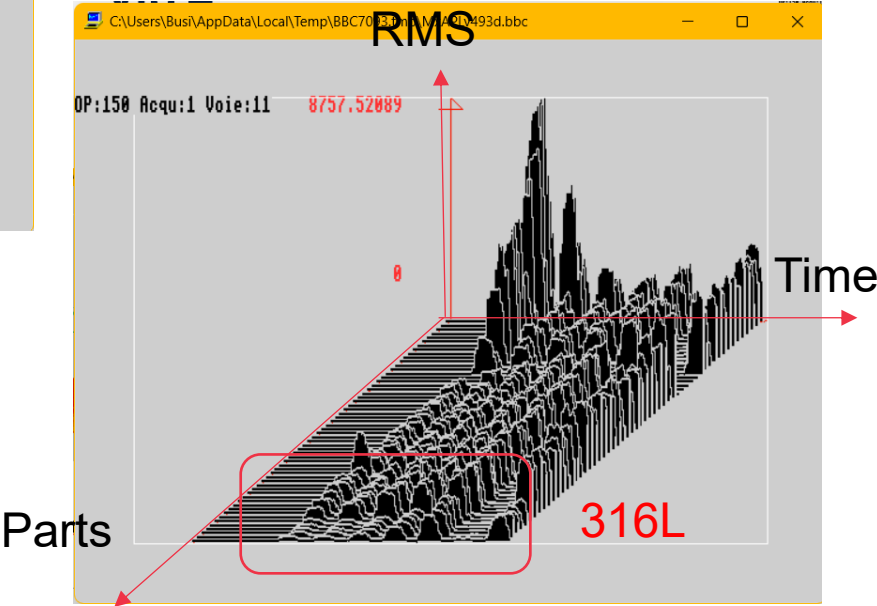
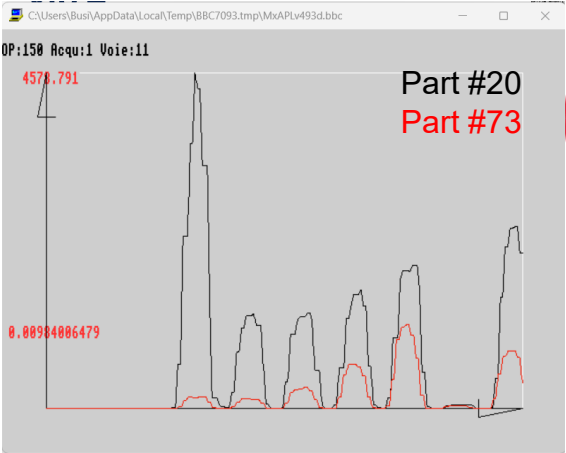
RMS, decreases a little when 316L grade

RMS



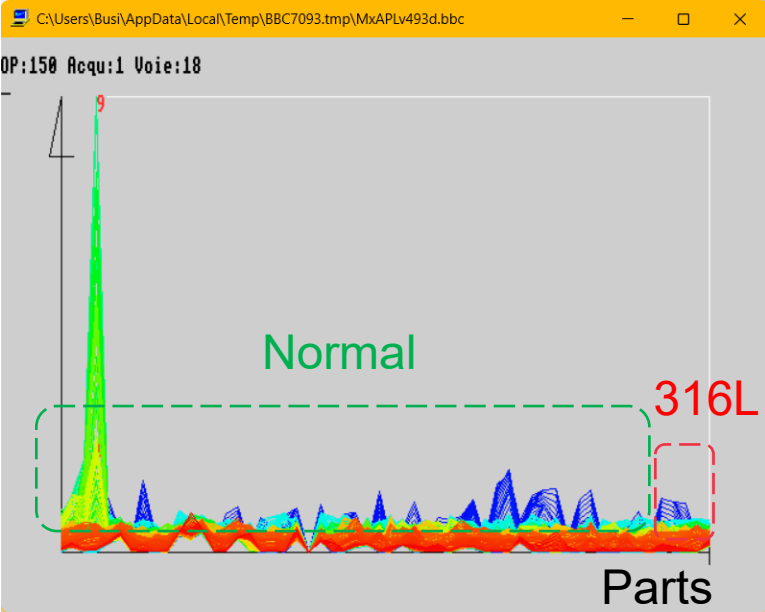
Before machining

In machining

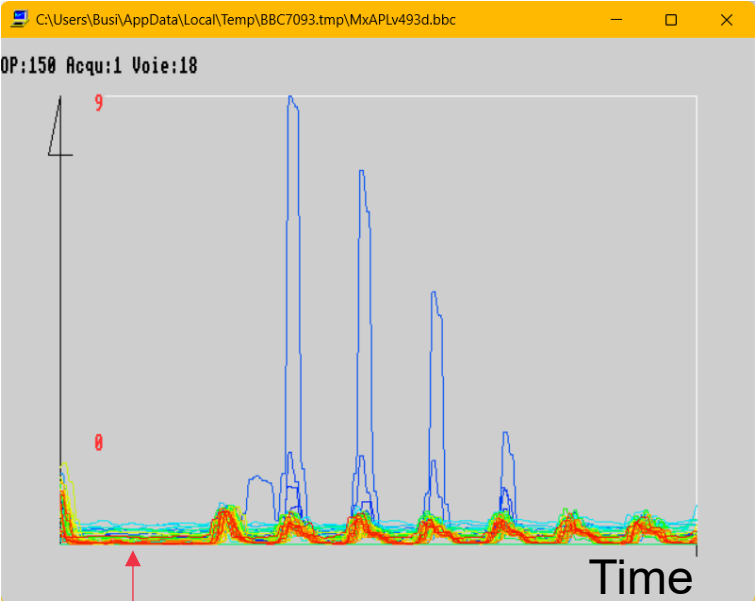


Audio : Vib 11

RMS

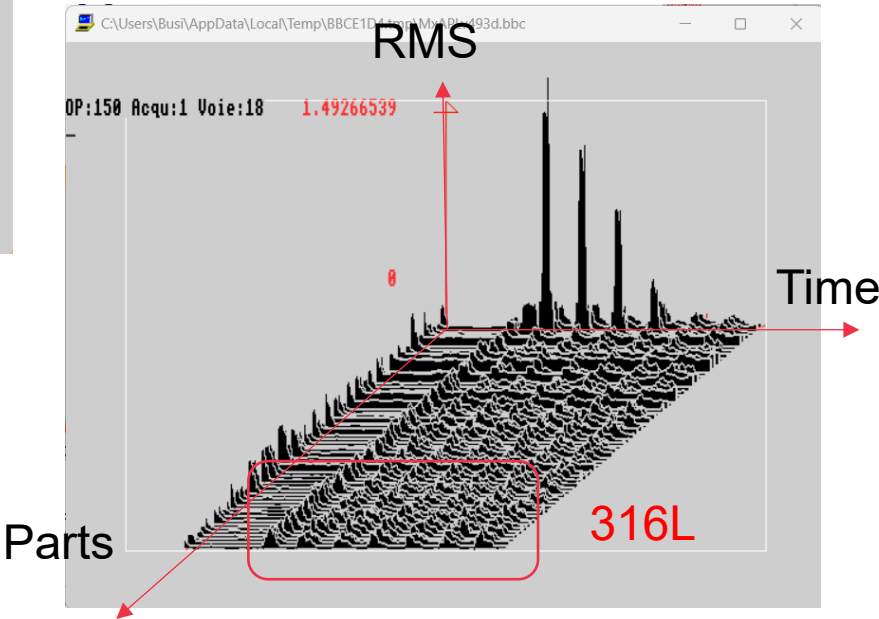
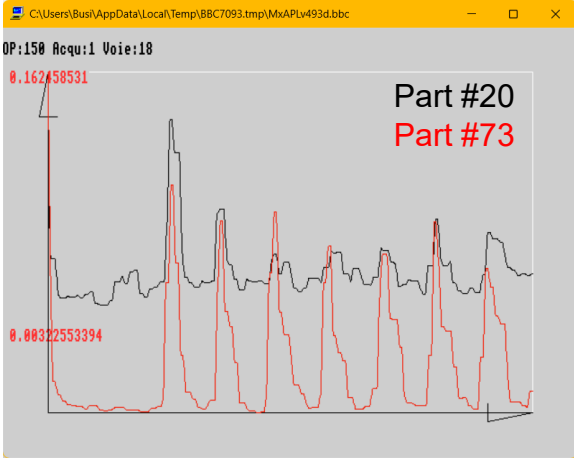


RMS



Before machining

In machining



Difficult to conclude !

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

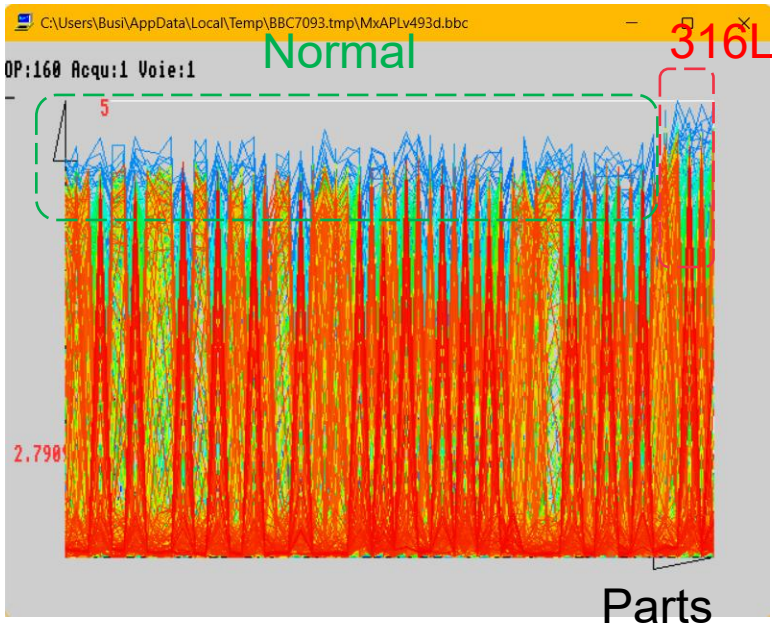
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<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
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<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" /> Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
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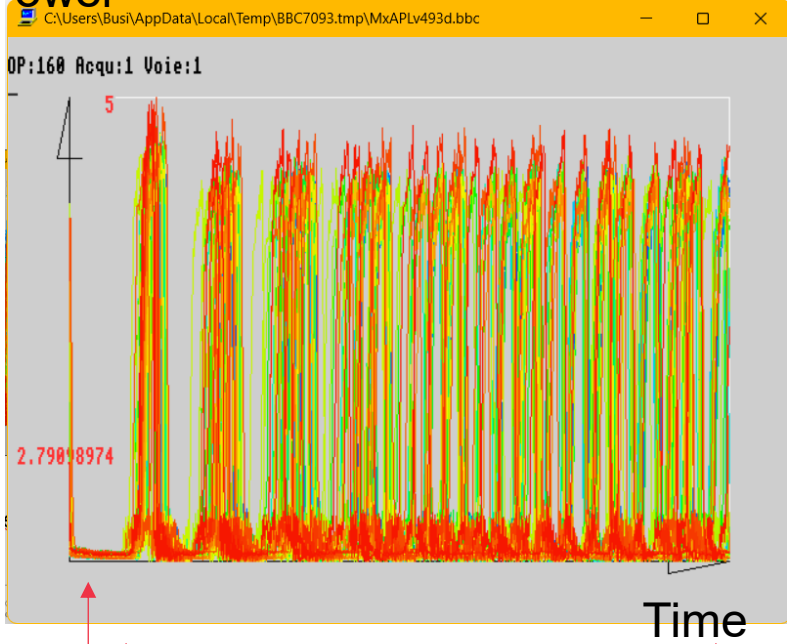
Machining OP 160

Spindle Power : Ai1

Power

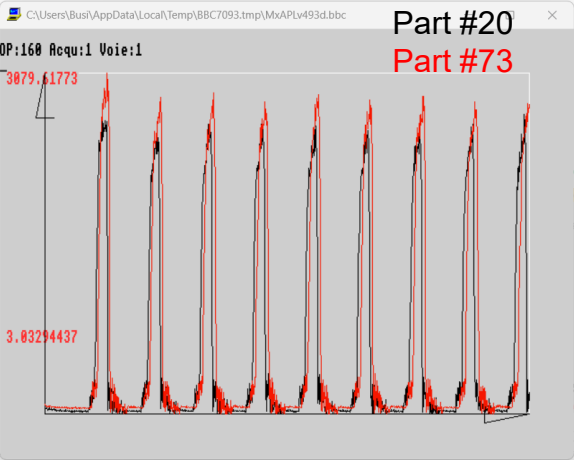


Power

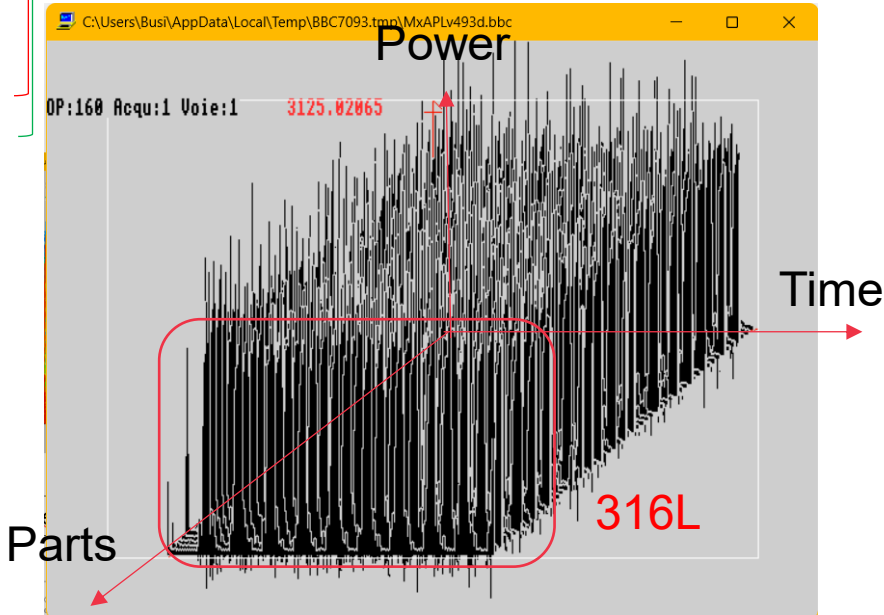


Before machining

In machining



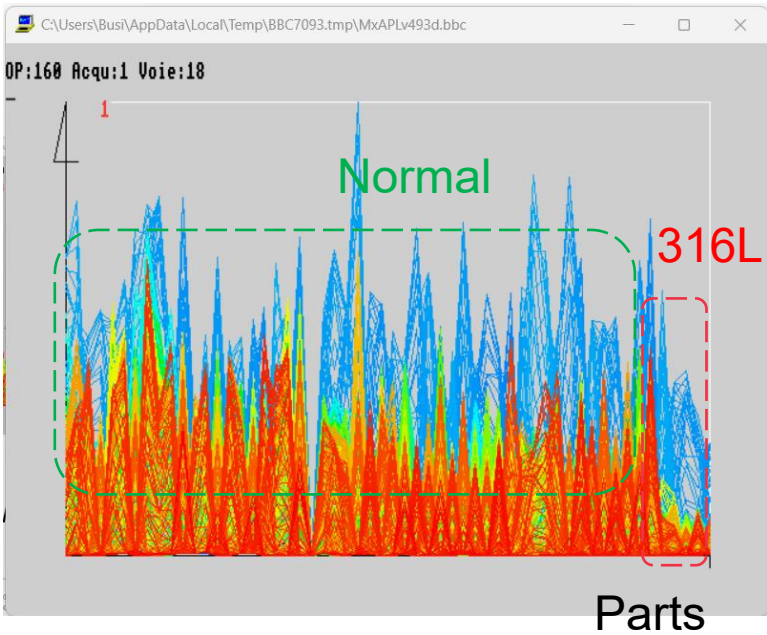
316L
Normal



Spindle power increases when 316L grade => OK

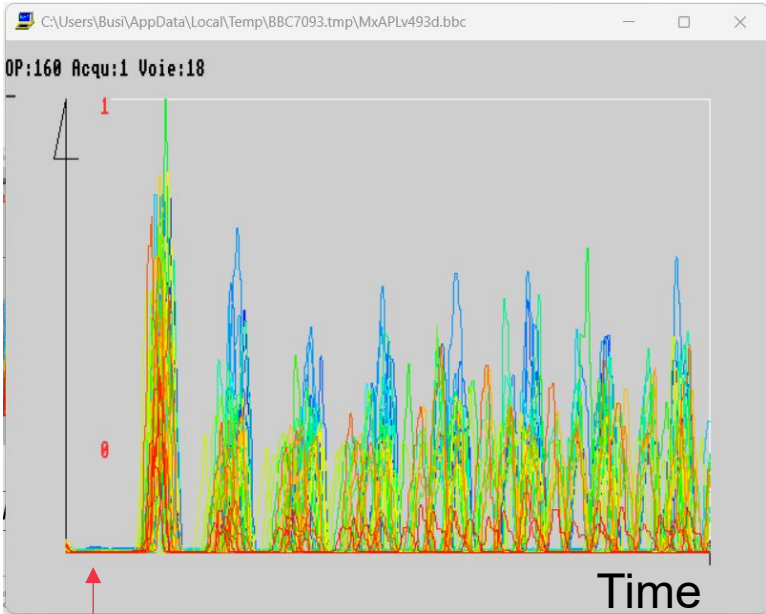
Audio : Vib 11

RMS



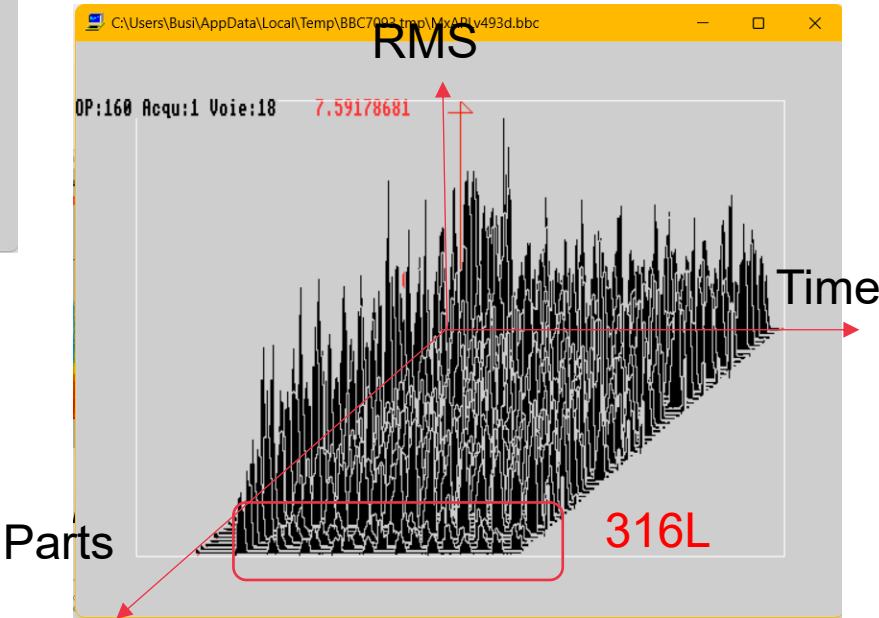
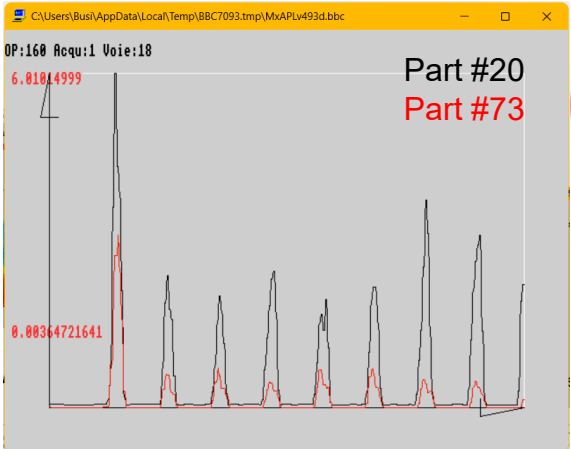
Audio RMS decreases when 316L grade

RMS



Before machining

In machining



Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

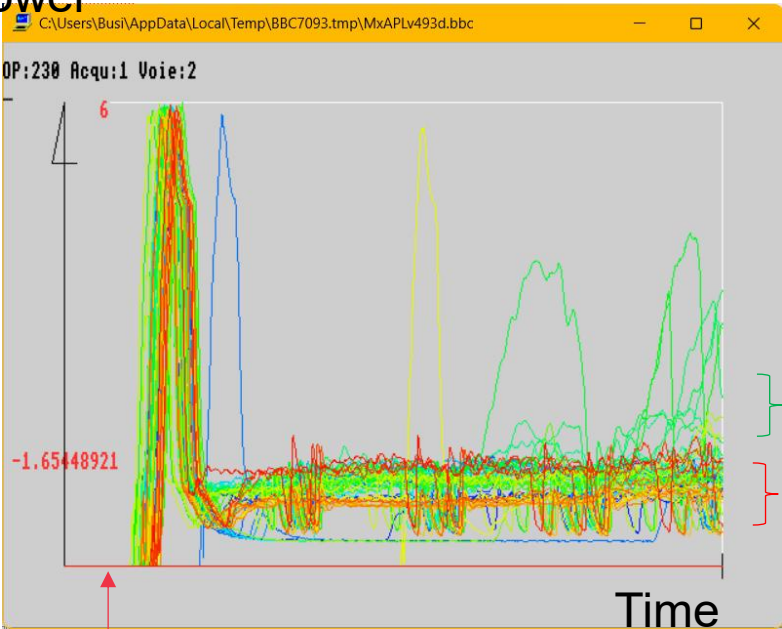
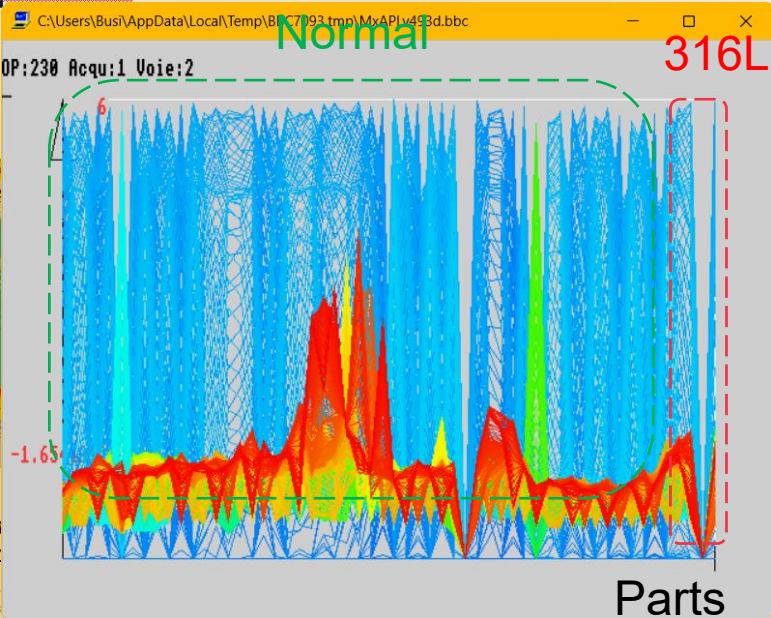
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<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 230

Spindle Power : Ai2

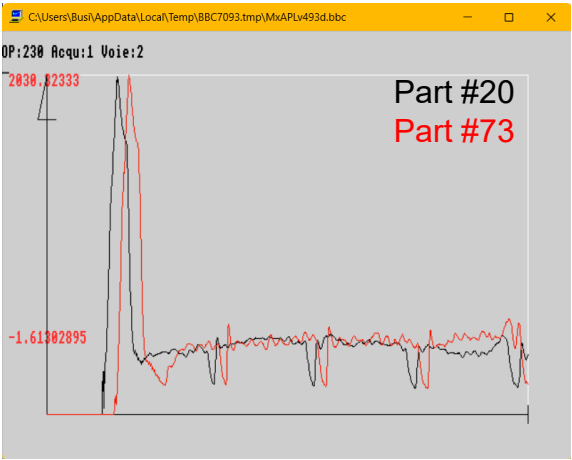
Power

Power



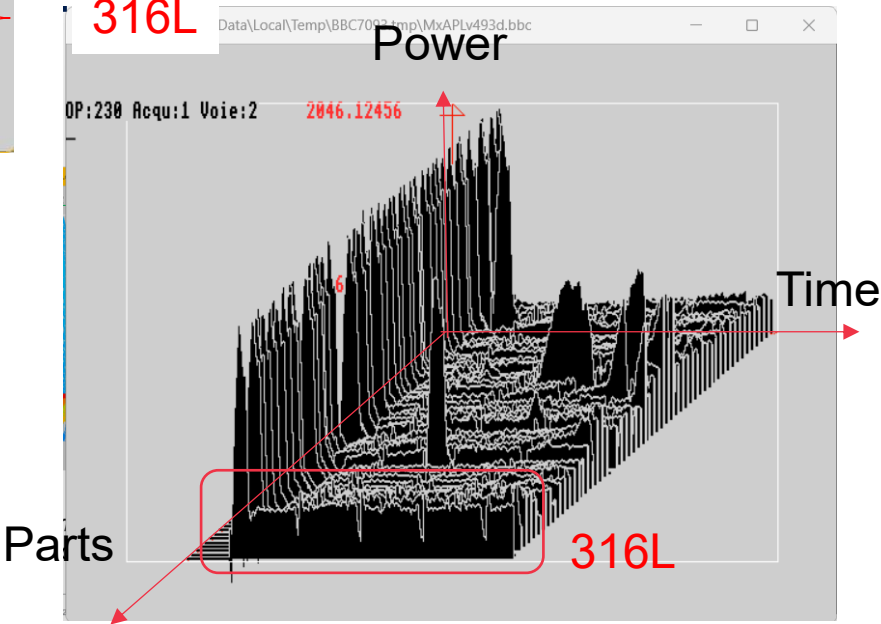
Before machining

In machining



Normal

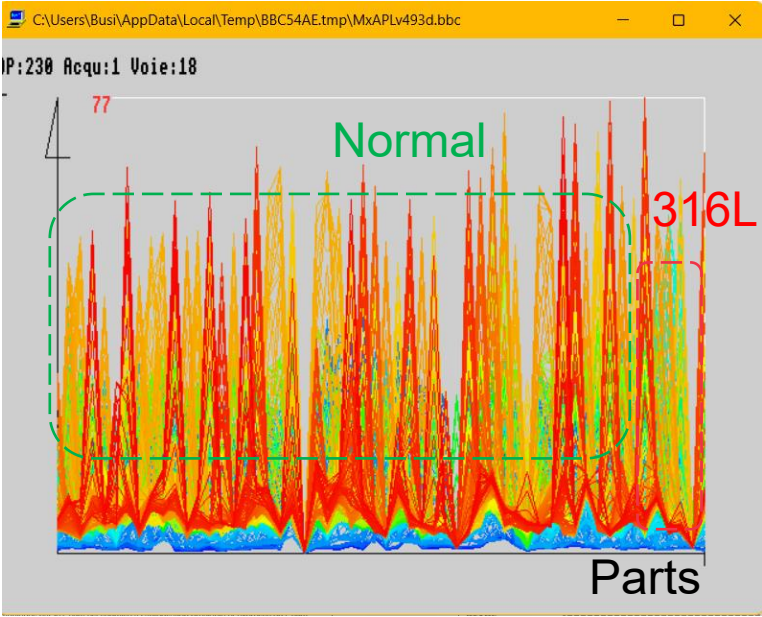
316L



Spindle power remains the same with 316L => OK

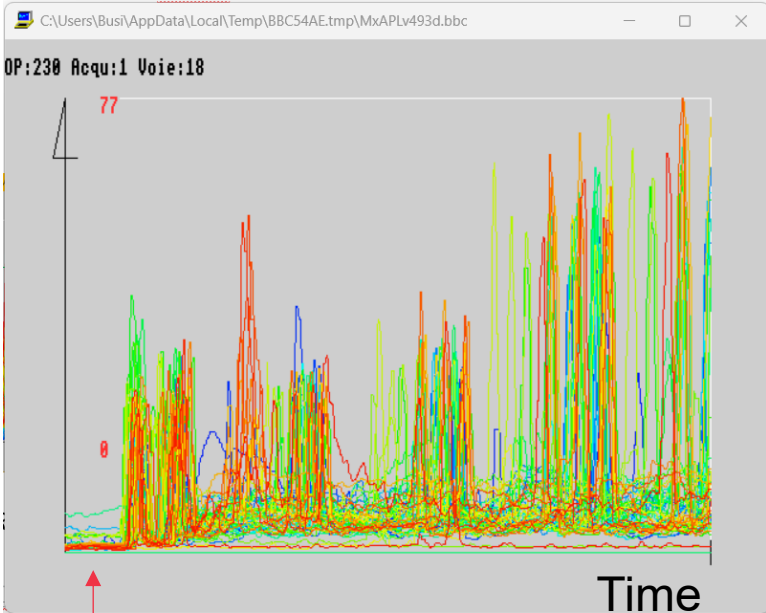
Audio : Vib 11

RMS



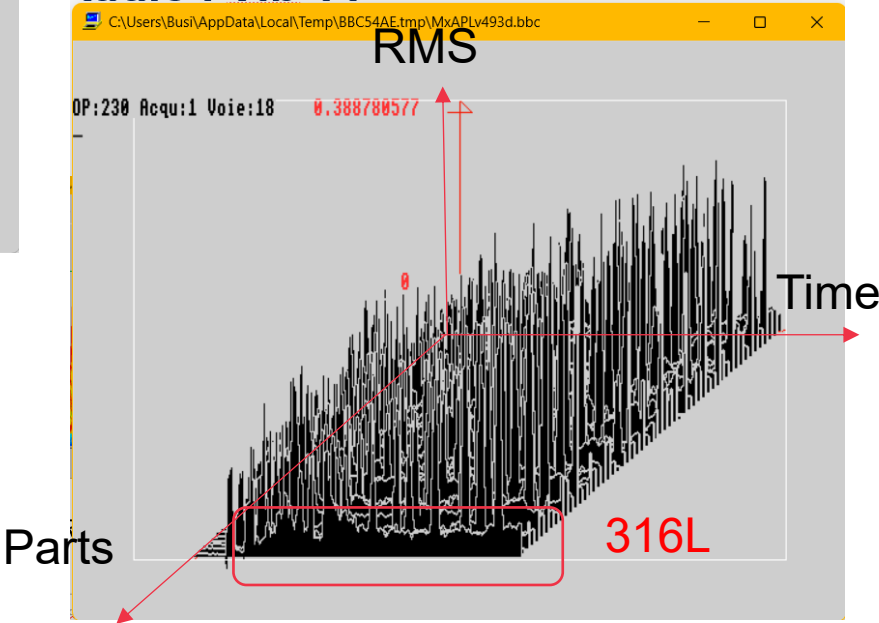
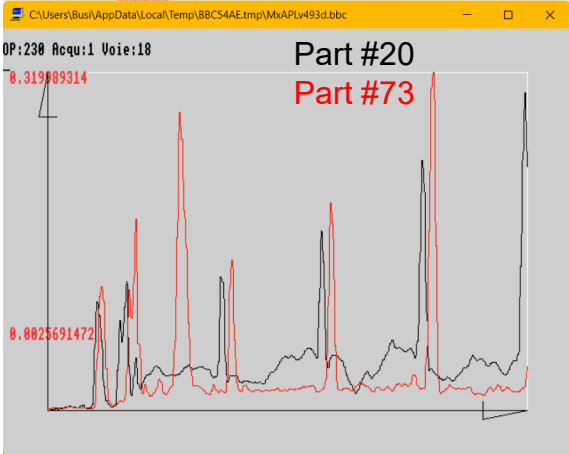
Audio RMS decreases a little when 316L grade

RMS



Before machining

In machining



Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

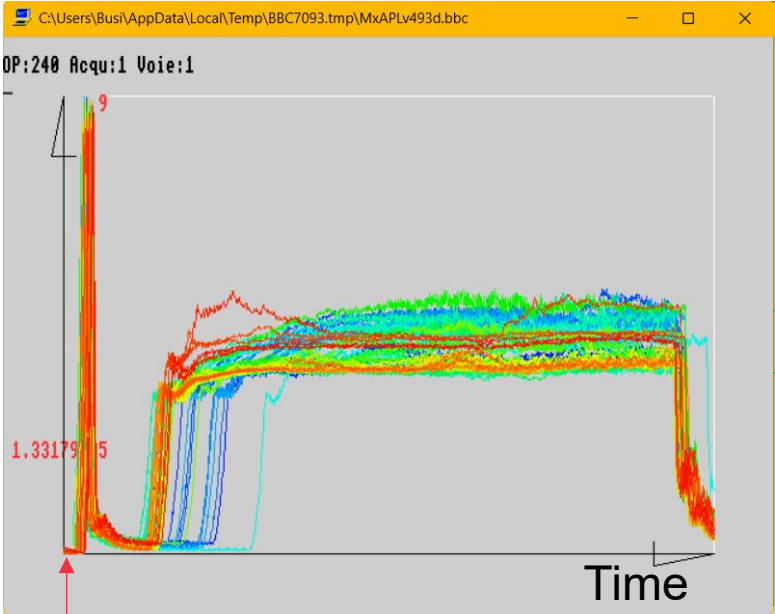
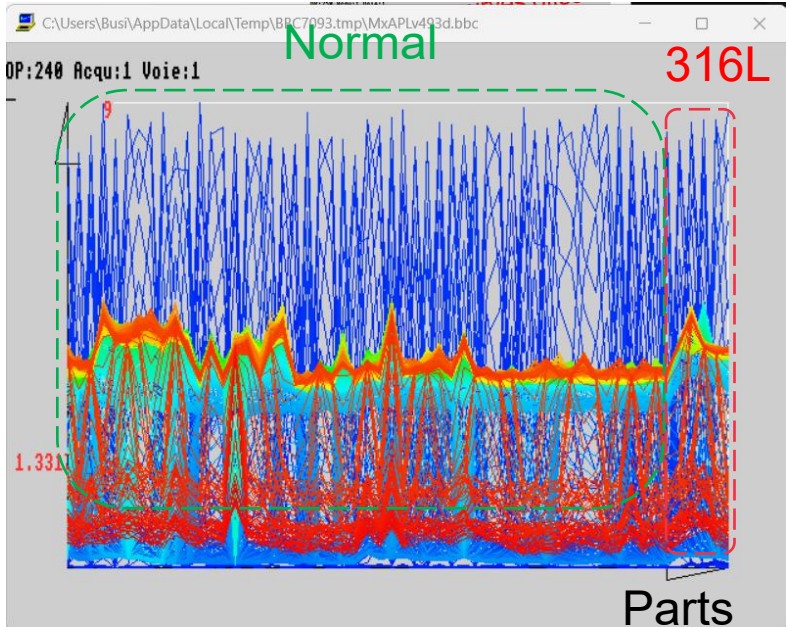
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<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
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Machining OP 240

Spindle Power : Ai1

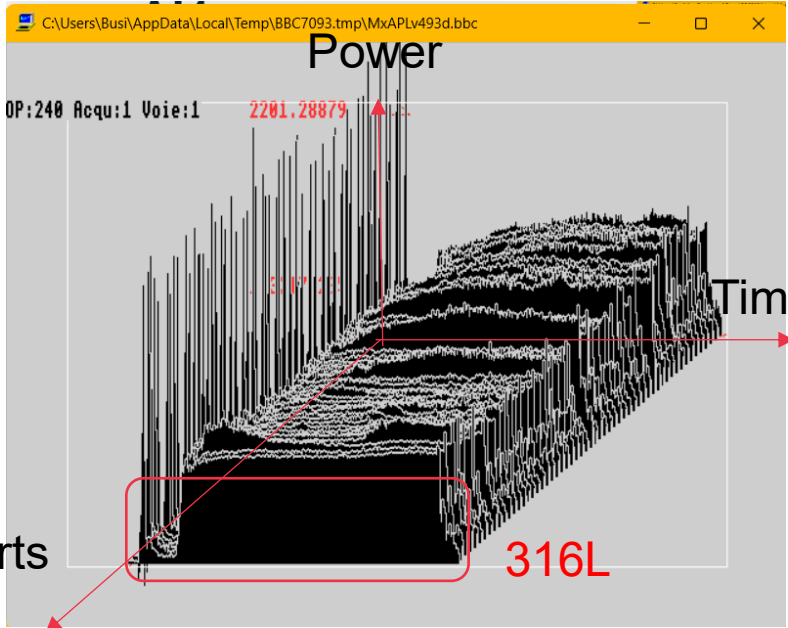
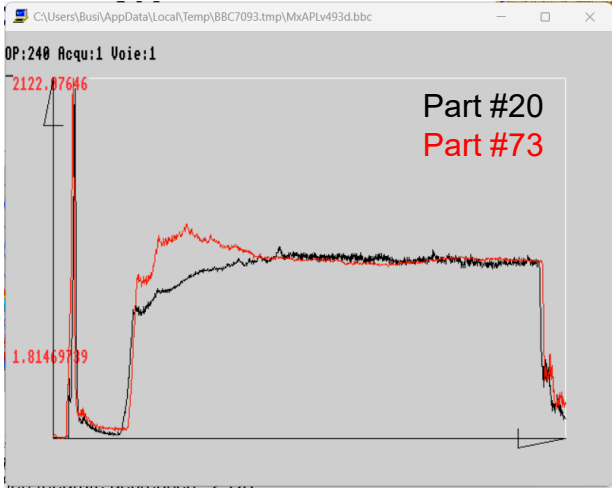
Power

Power



Before machining

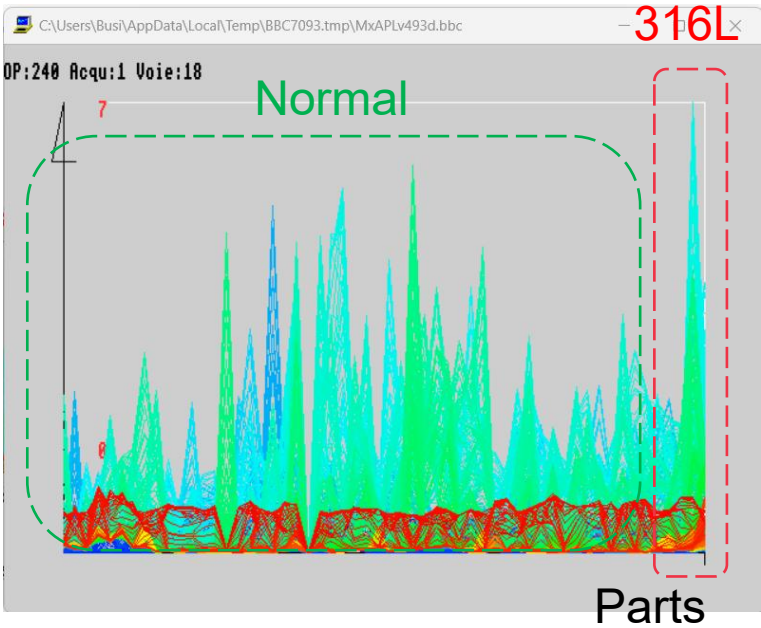
In machining



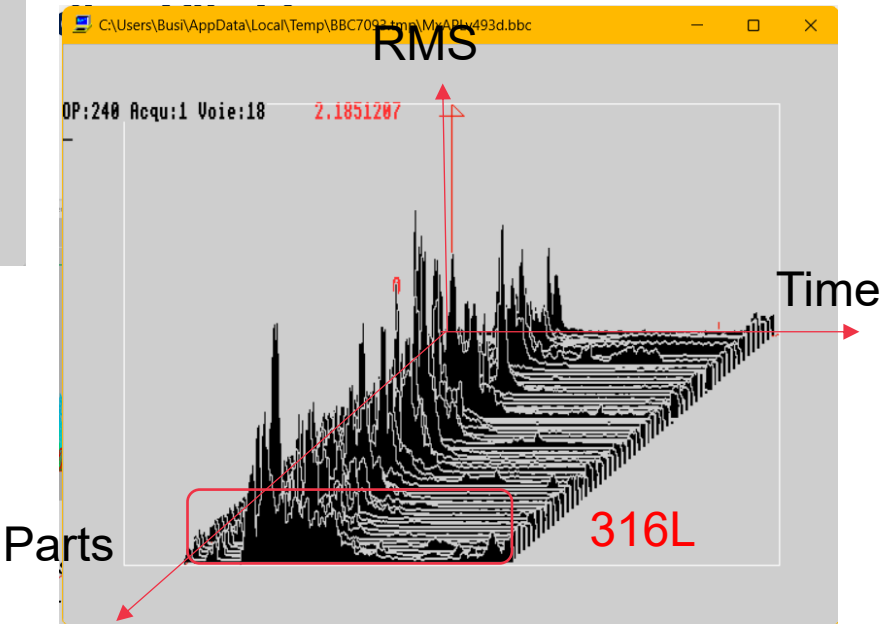
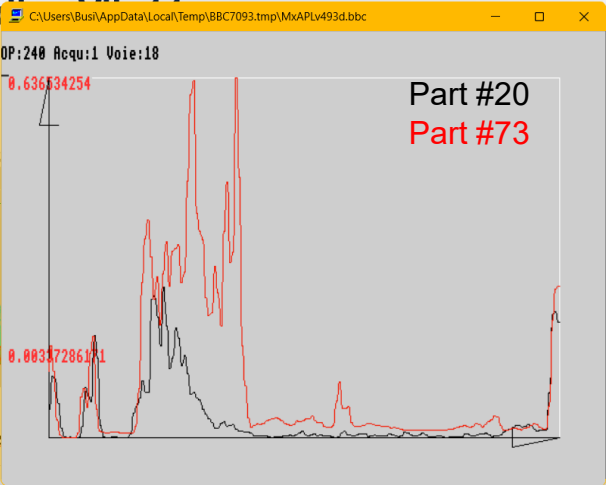
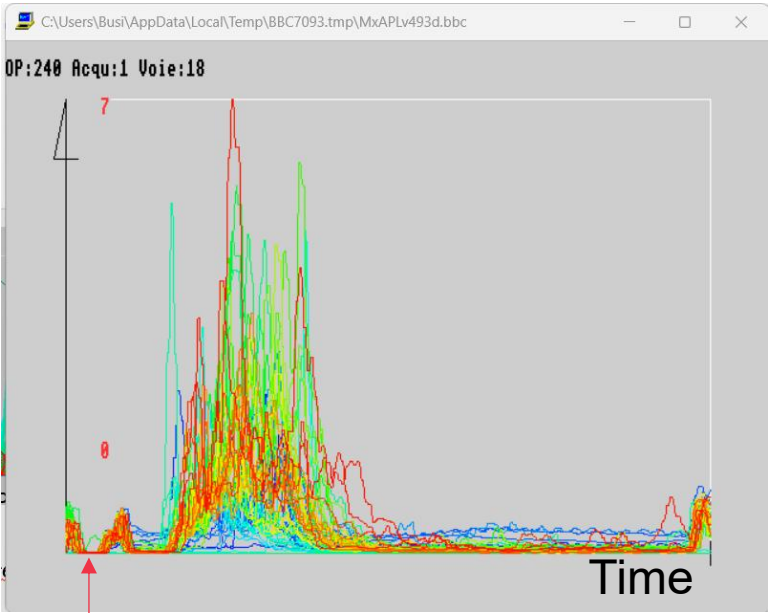
Rem : Temporal shifts sometimes !
Spindle power increases when 316L grade => OK

Audio : Vib 11

RMS



RMS



Audio RMS increases when 316L grade

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

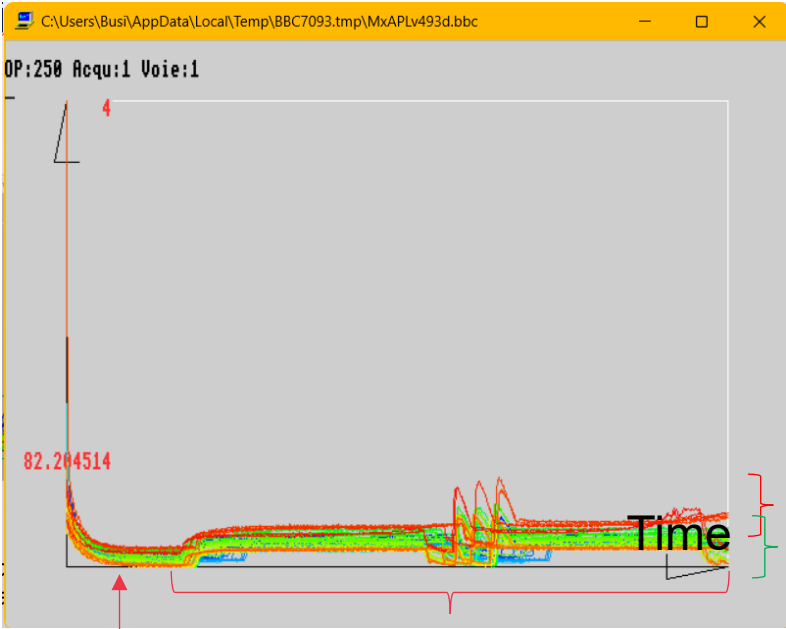
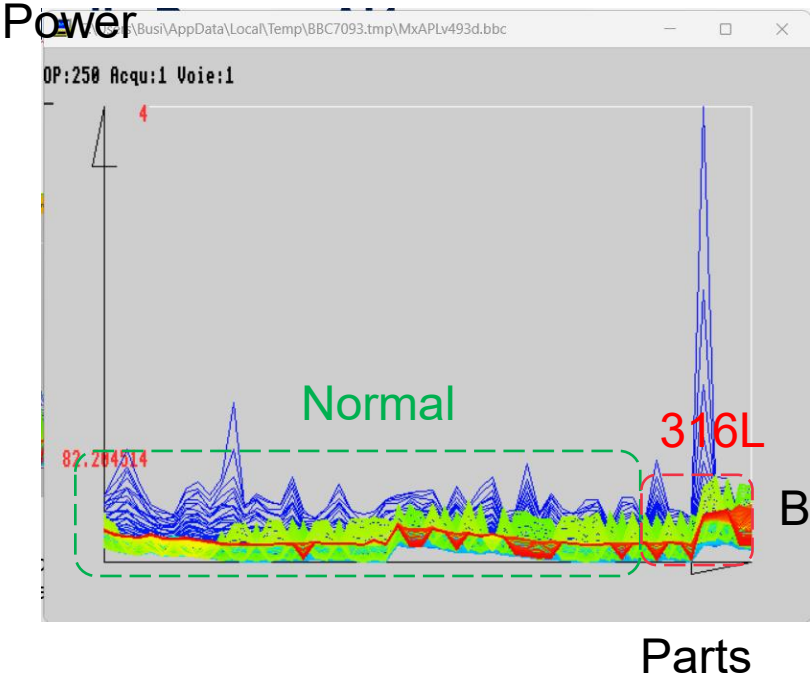
#18 : Vib11 (audio micro)

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<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" /> Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

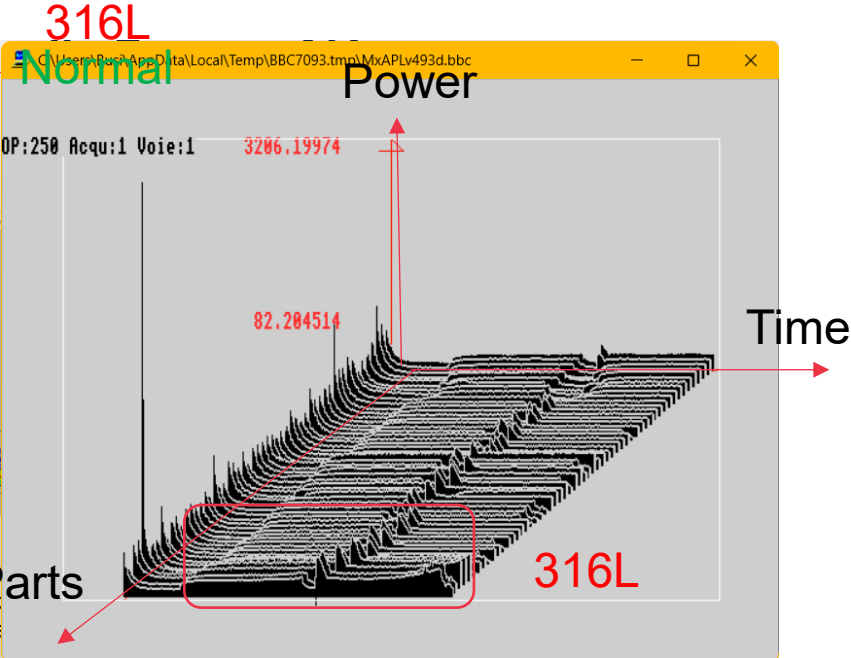
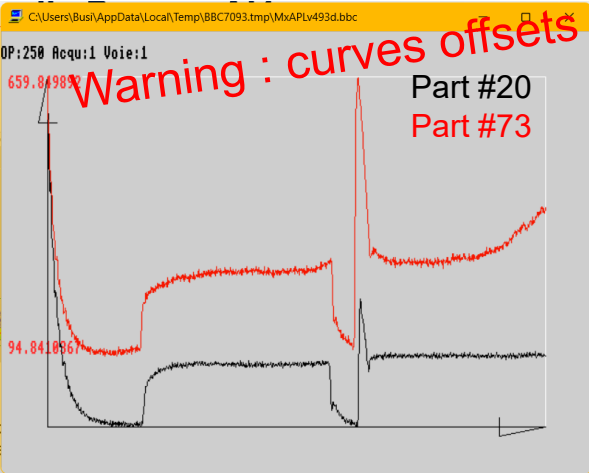
Machining OP 250

Spindle Power : Ai1



Before machining

In machining



Spindle power increases when 316L grade => OK

Vib : Vib3



Artefact => To Do again

Audio : Vib 11



Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

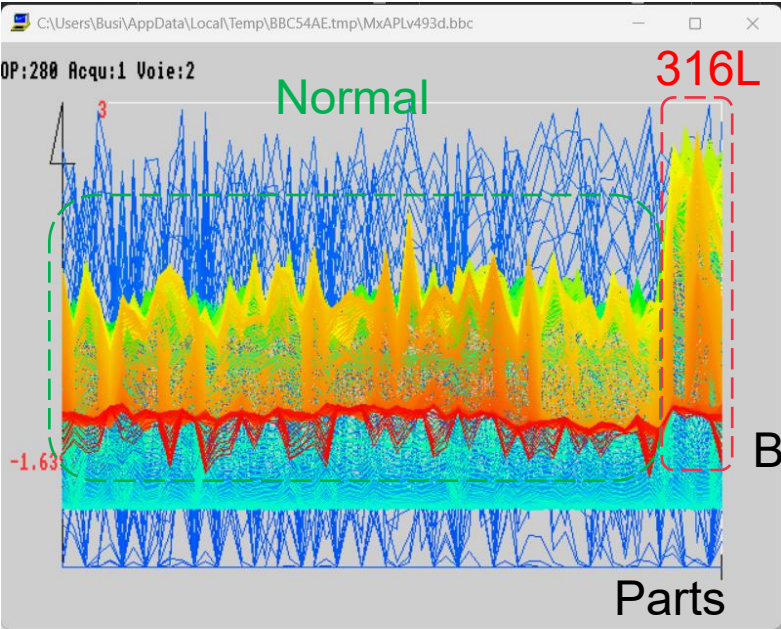
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Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

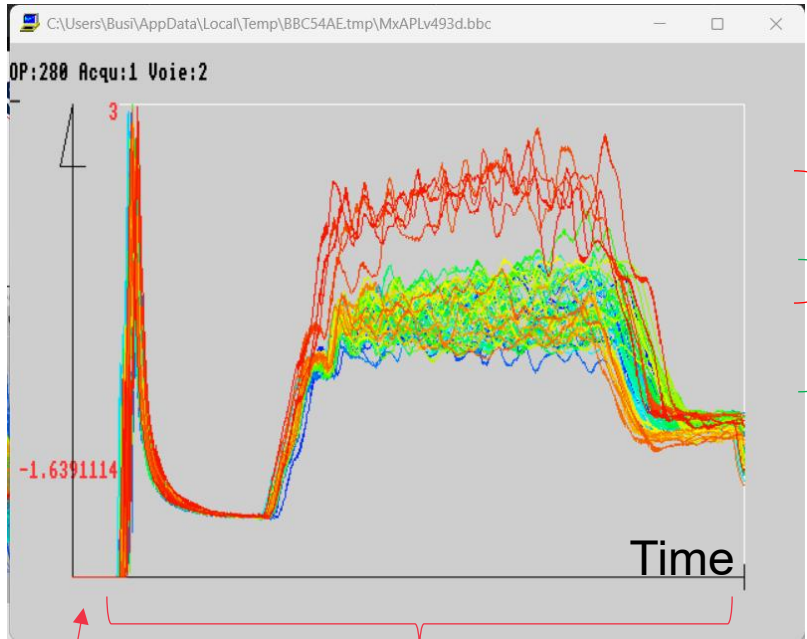
Machining OP 280

Spindle Power : Ai2

Power



Power

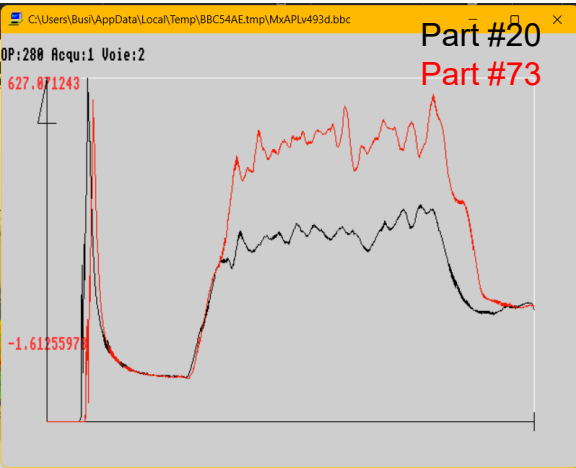


Before machining

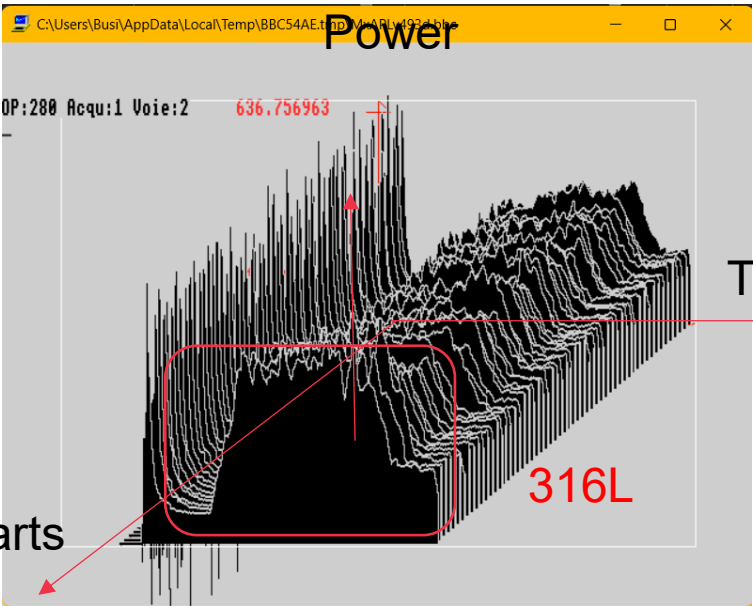
In machining

316L

Normal



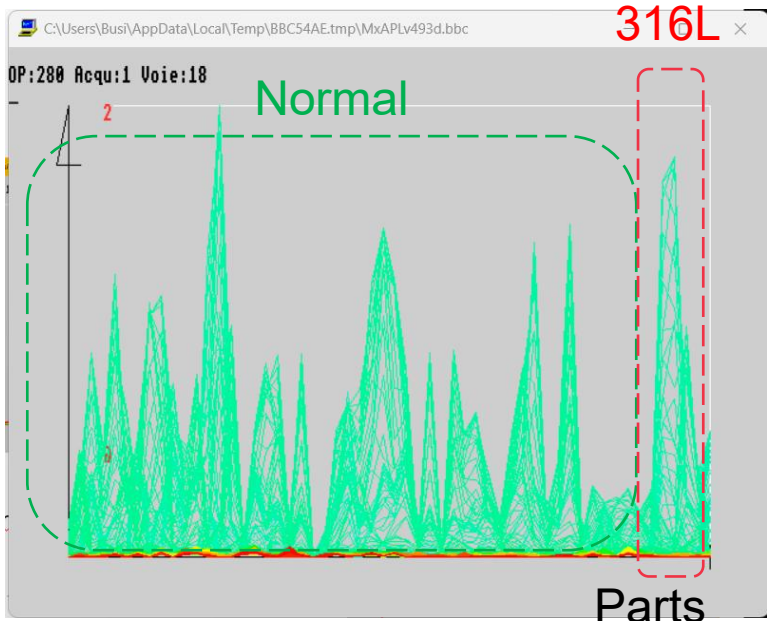
Power



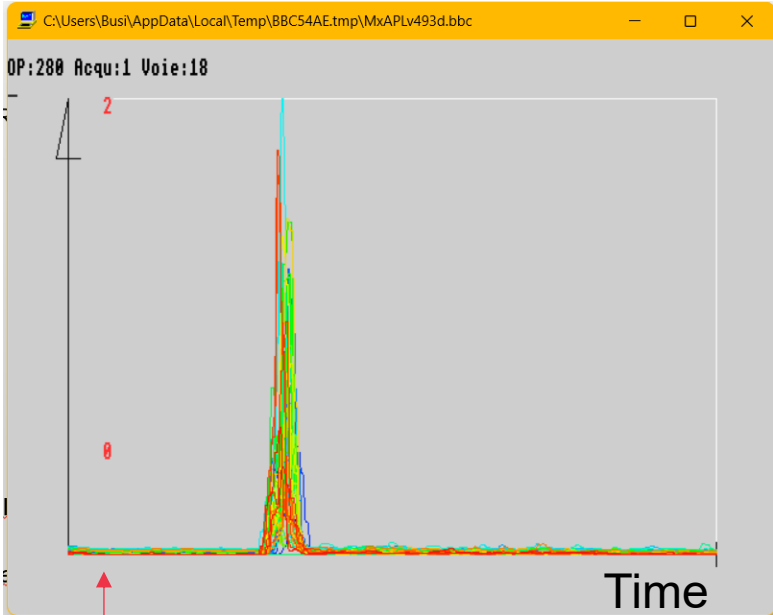
Spindle power increases when 316L grade => OK

Audio : Vib 11

RMS

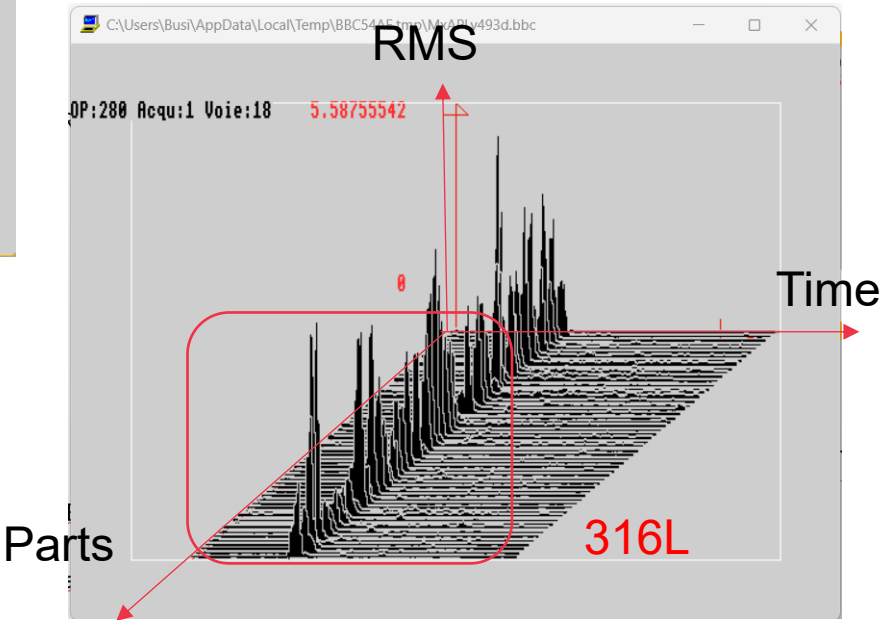
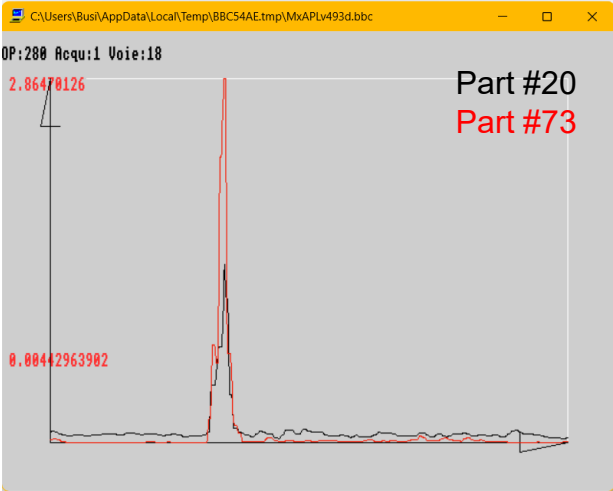


RMS



Before machining

In machining



Audio RMS seems to increase when 316L grade

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

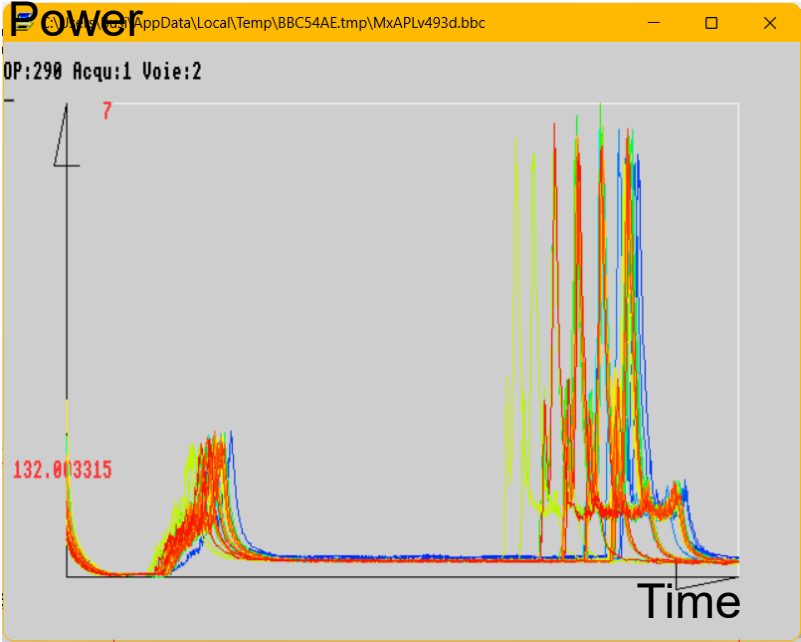
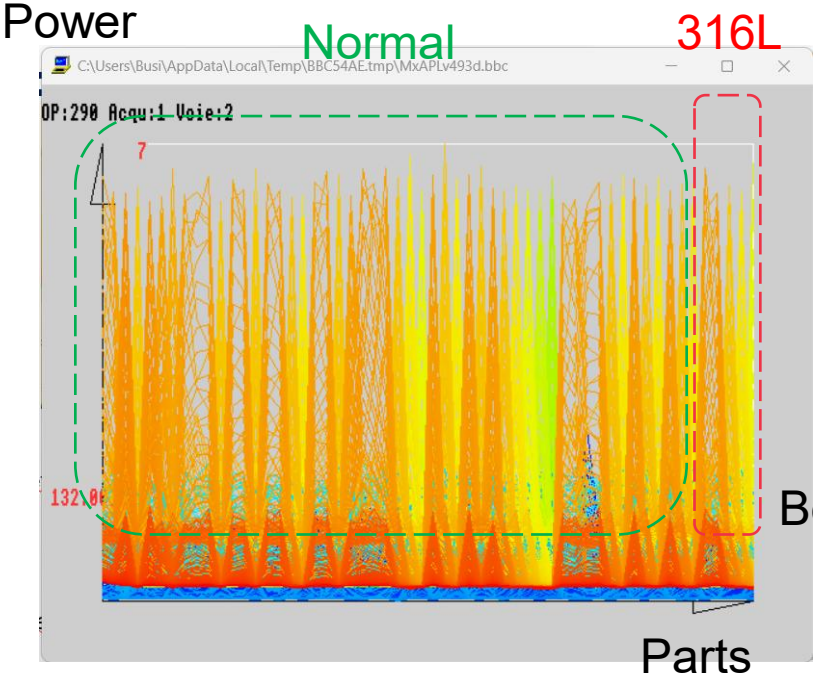
#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.099999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
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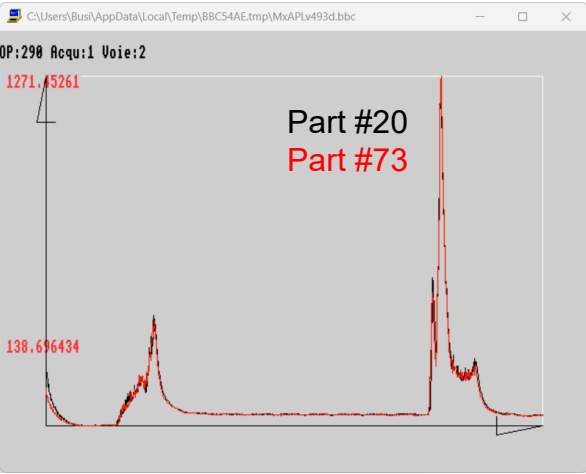
Machining OP 290

Spindle Power : Ai2



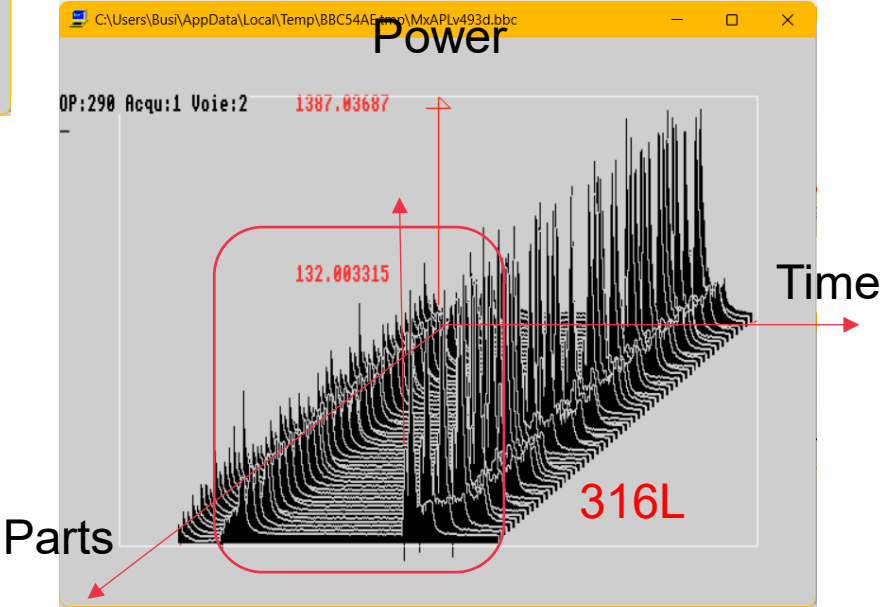
Before machining

In machining



Normal

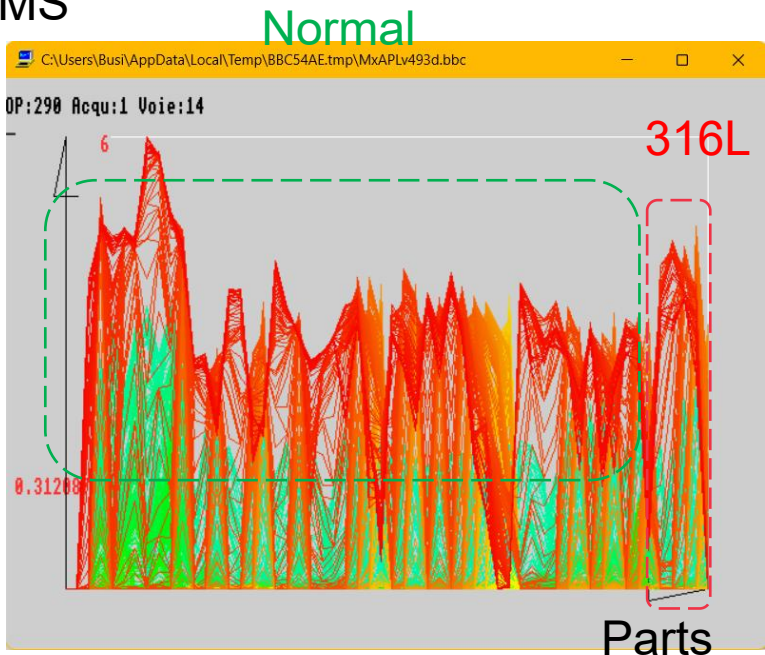
316L



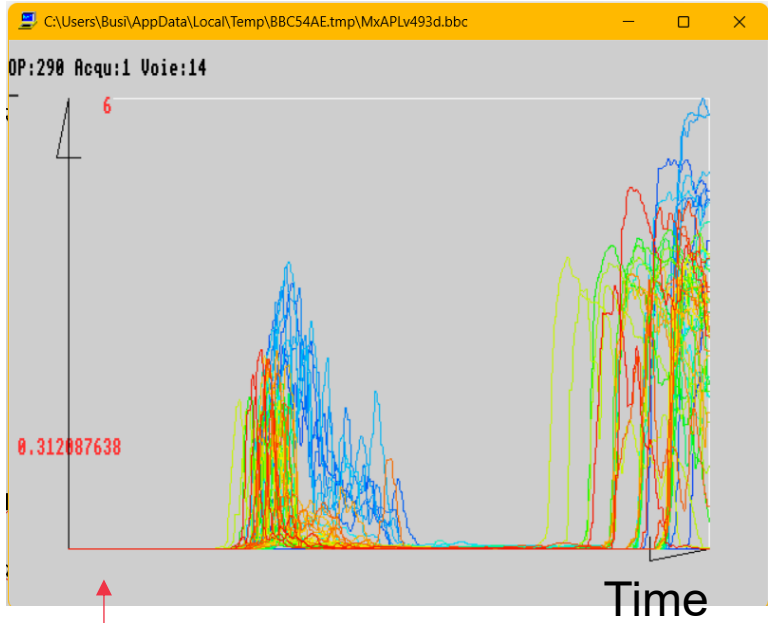
Spindle power remains the same with 316L grade ?

Vib : Vib 7

RMS

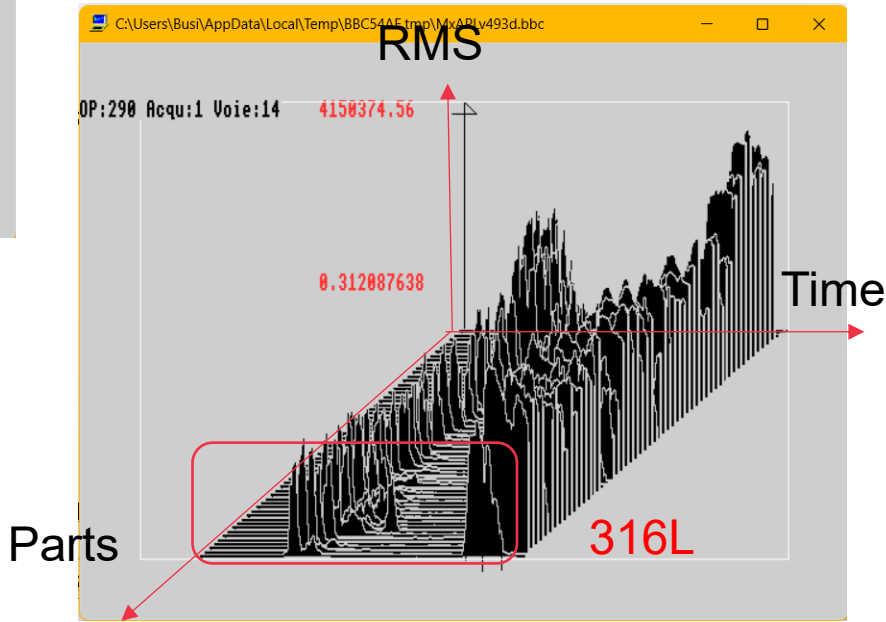
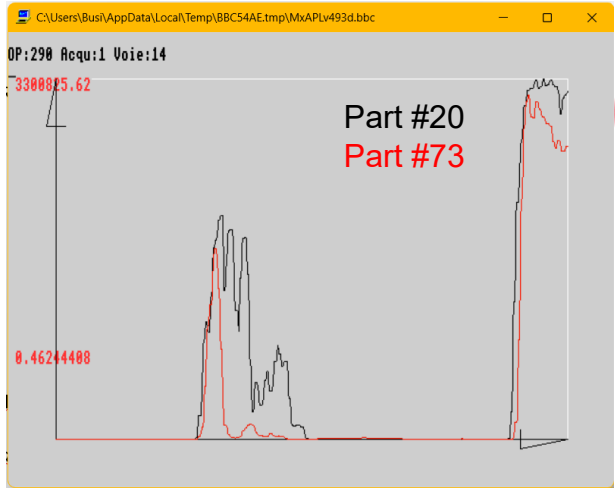


RMS



Before machining

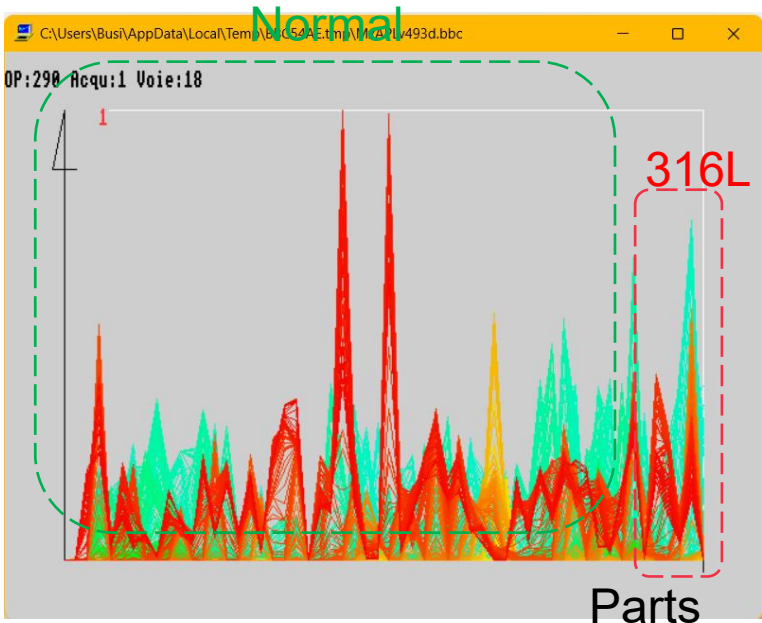
In machining



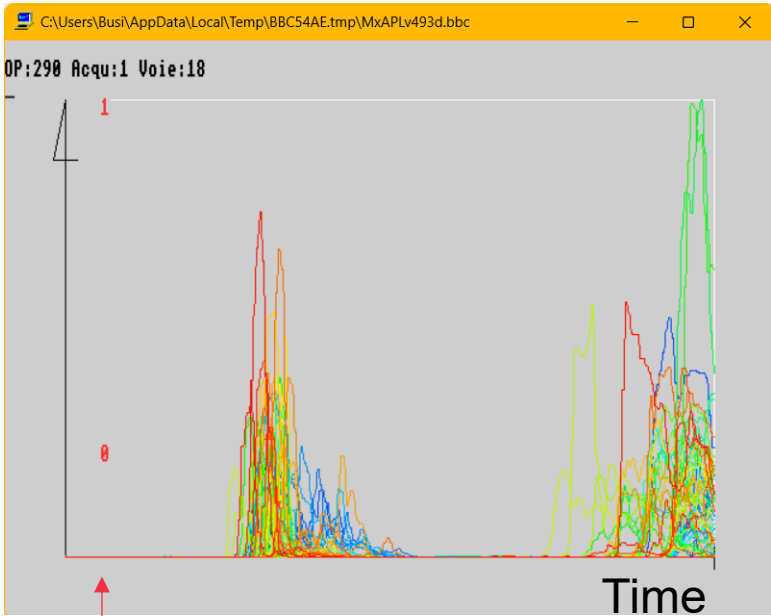
RMS remains the same with 316L grade

Audio : Vib 11

RMS

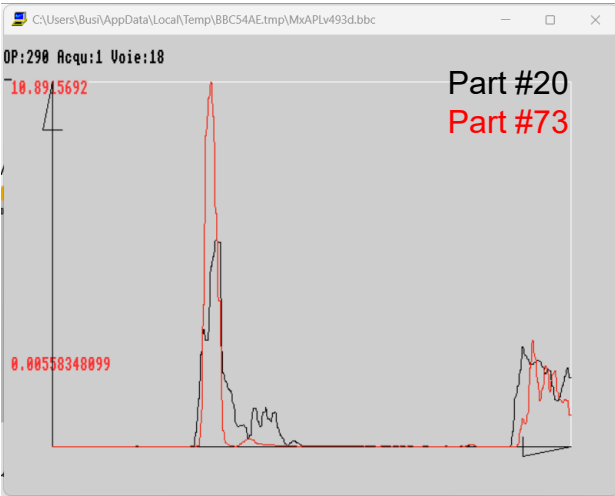


RMS

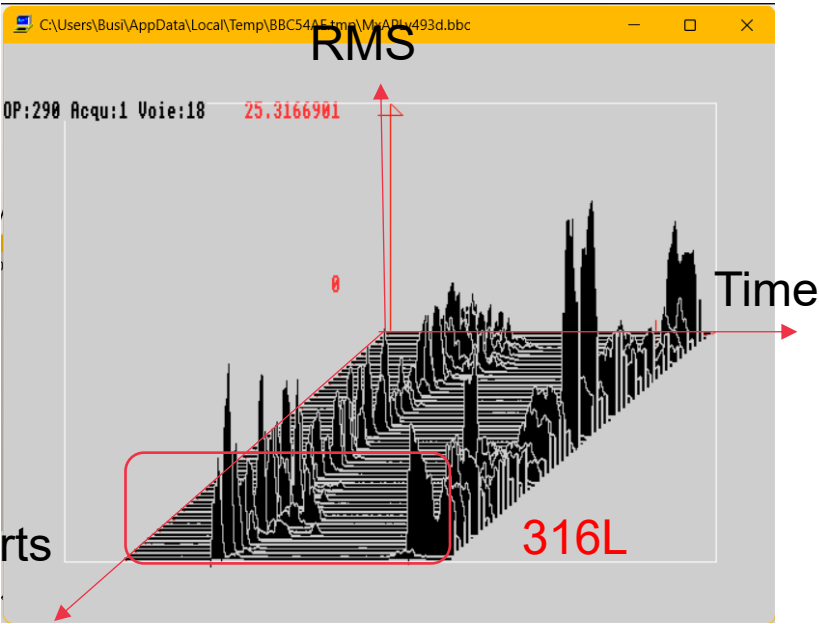


Before machining

In machining



RMS



Audio RMS seems to increase when 316L grade ?

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
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<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
Le 26/01/26
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```

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<Operation id="250" description="AL ESAGE DIAM 10" monitor="true" tool="T02" Color="0;255;0" />
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<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 311

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

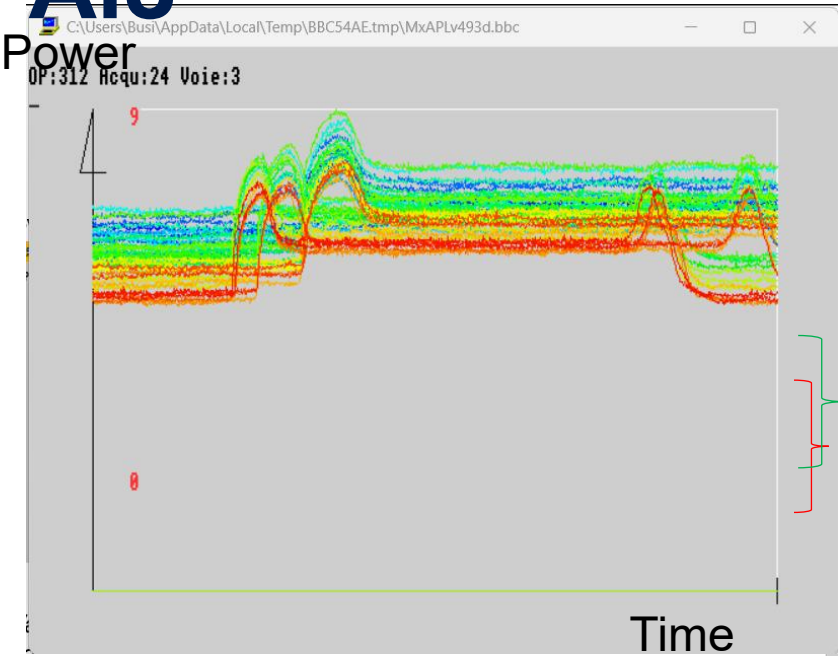
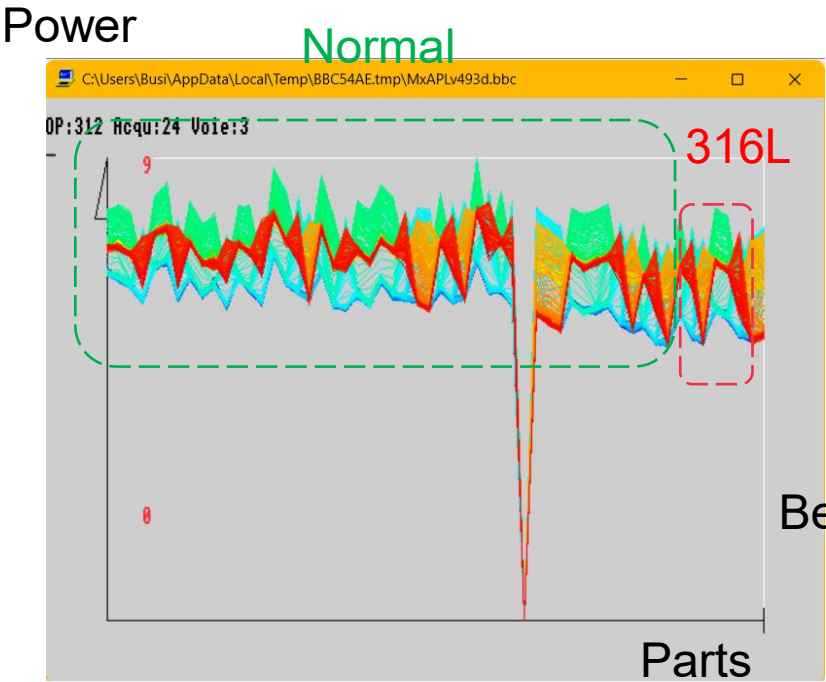
```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" /> Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 312

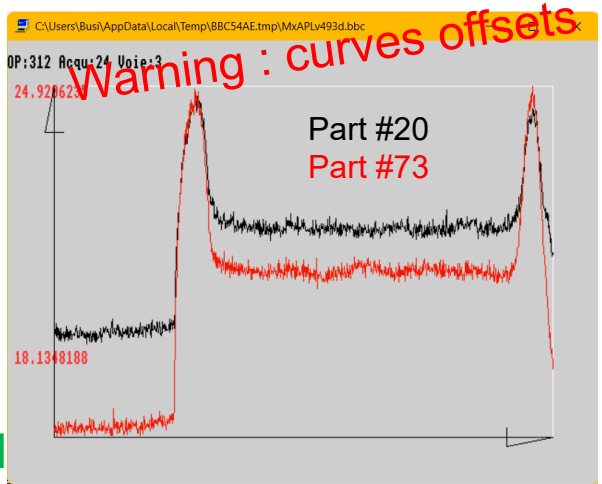
(last pass #24)

Spindle Power : Ai3



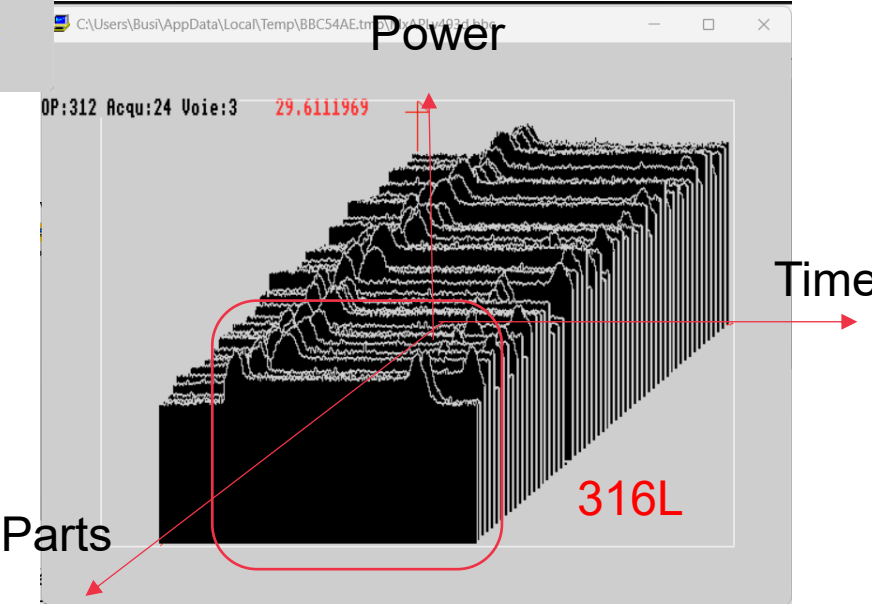
Before machining

In machining



Normal

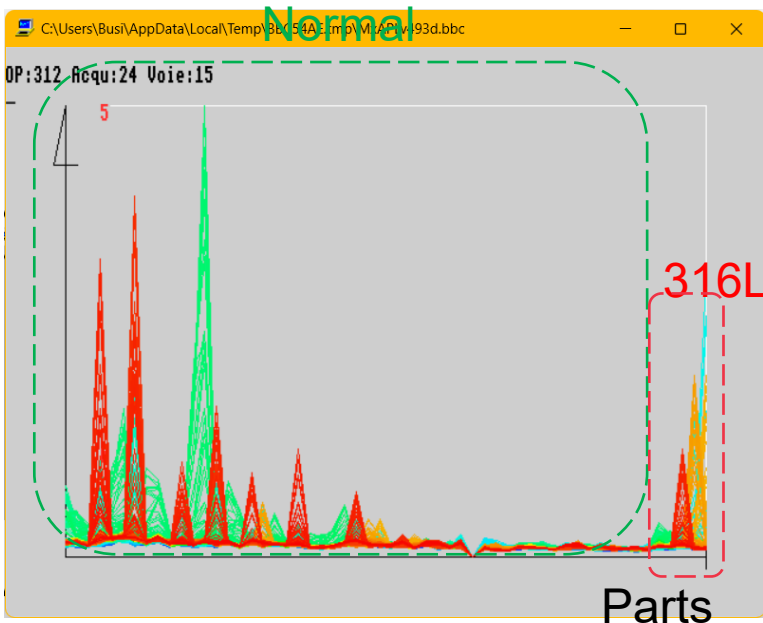
316L



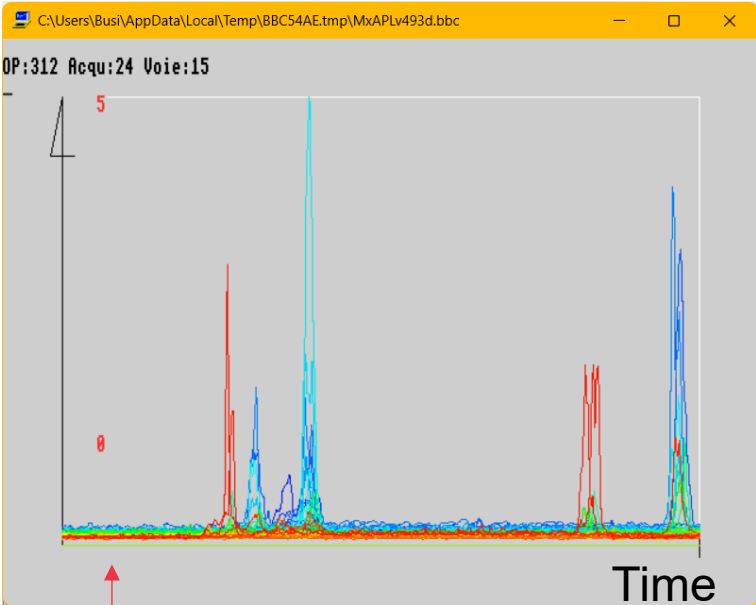
Warning to the vertical offset : Spindle power increases when 316L grade => OK

Vib : Vib 8

RMS

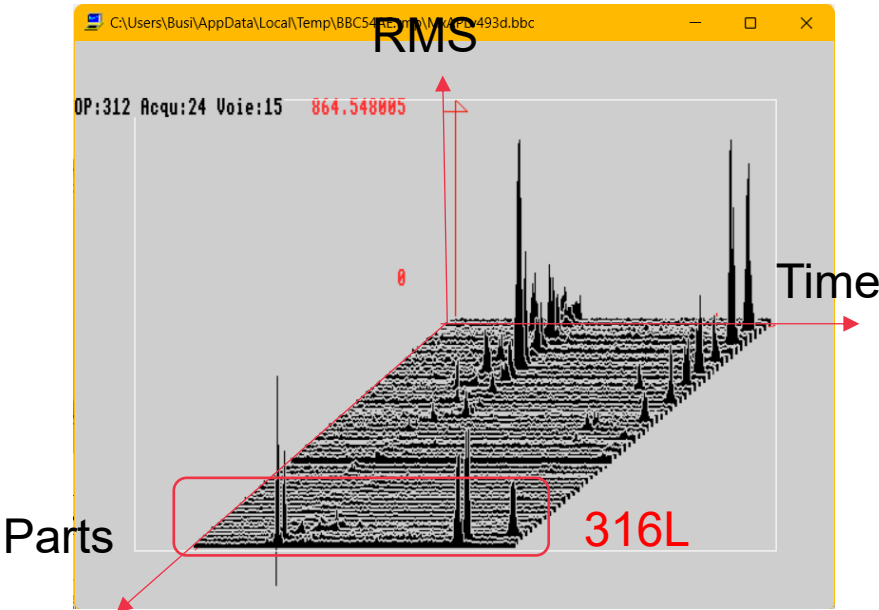
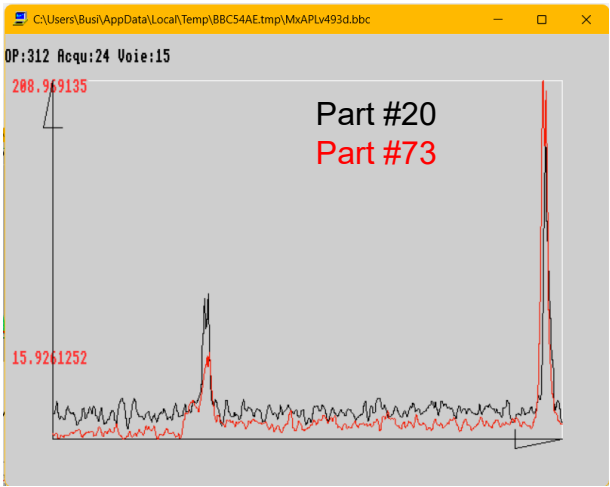


RMS



Before machining

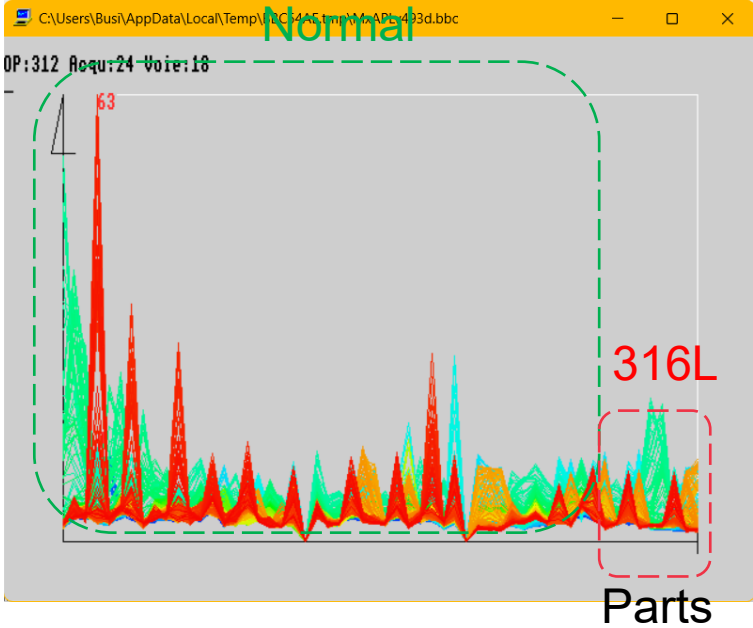
In machining



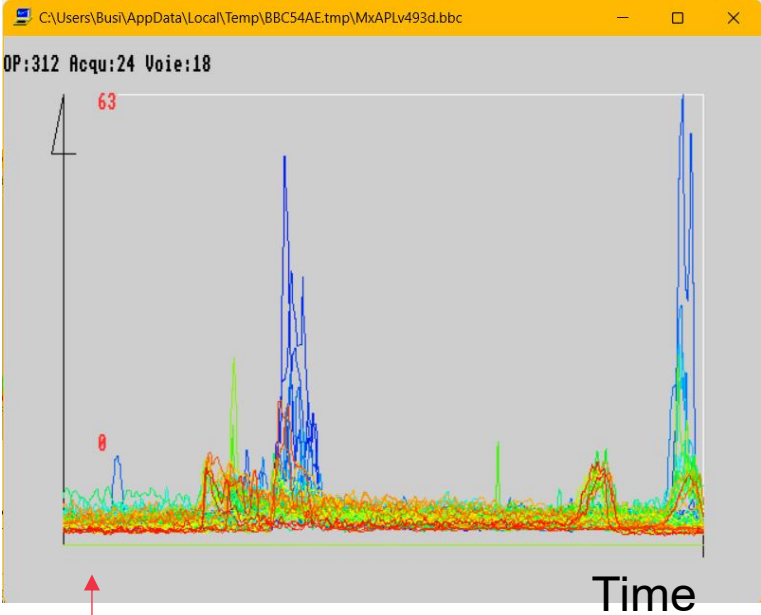
RMS increases when 316L grade compared to the previous normal part produced ?

Audio : Vib 11

RMS



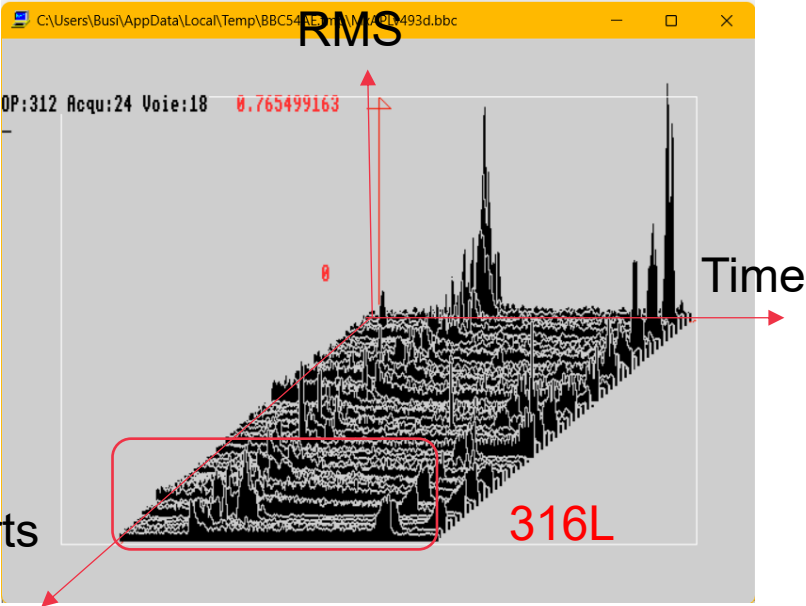
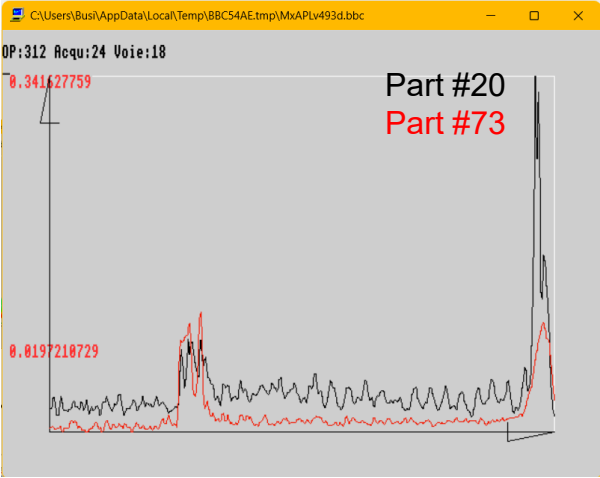
RMS



Before machining

In machining

Audio RMS, seems to decrease when 316L grade



Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="250" description="ALÉSAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="260" description="ALÉSAGE DIAM 11" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="ALÉSAGE DIAM 12" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="280" description="ALÉSAGE DIAM 13" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="290" description="ALÉSAGE DIAM 14" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 313

(last pass #24)

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

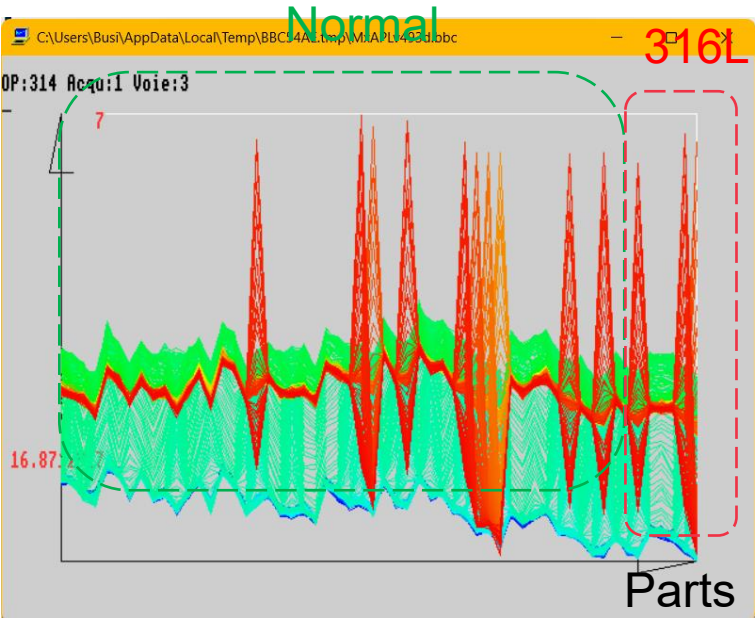
```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

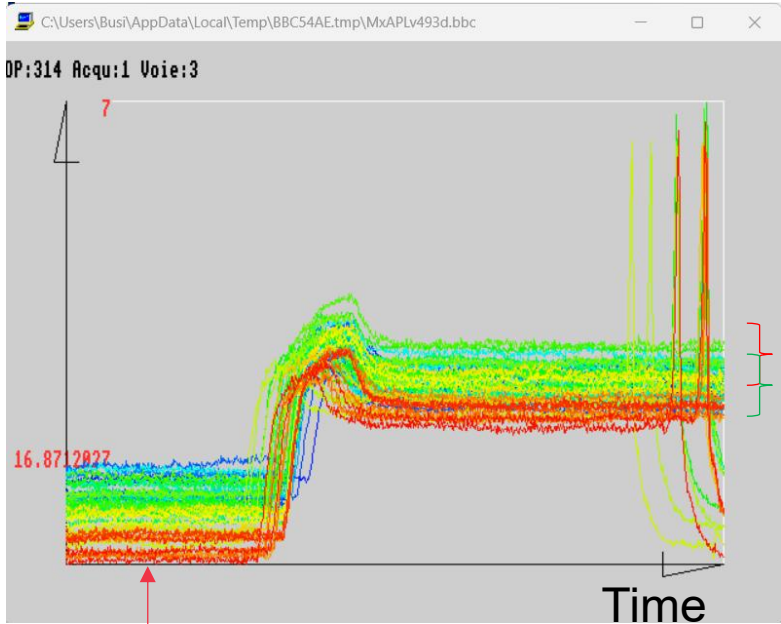
Machining OP 314

Spindle Power : Ai3

Power

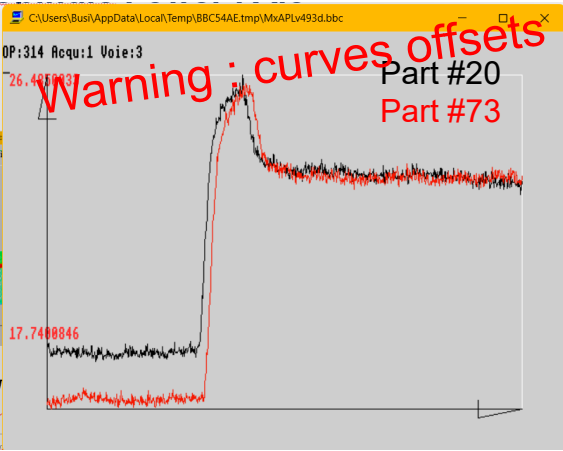


Power

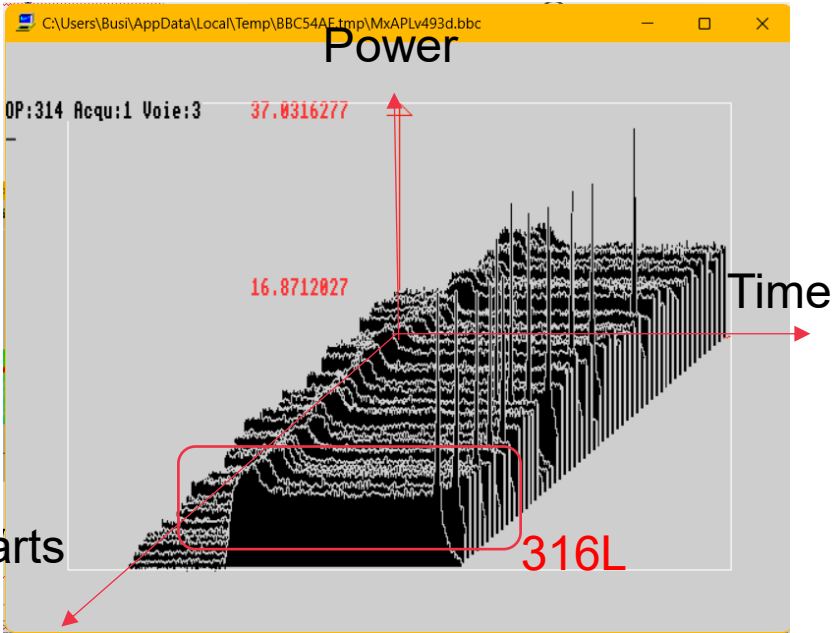


Before machining

In machining



316L
Normal

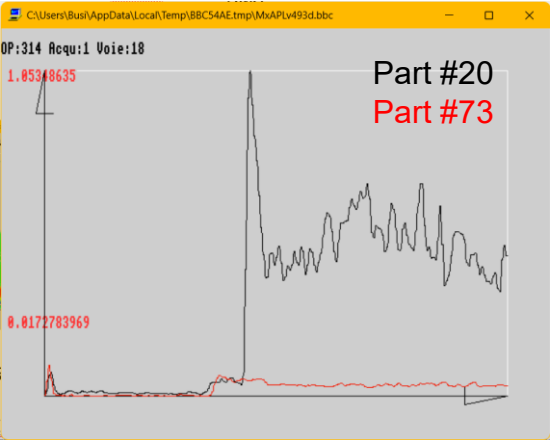
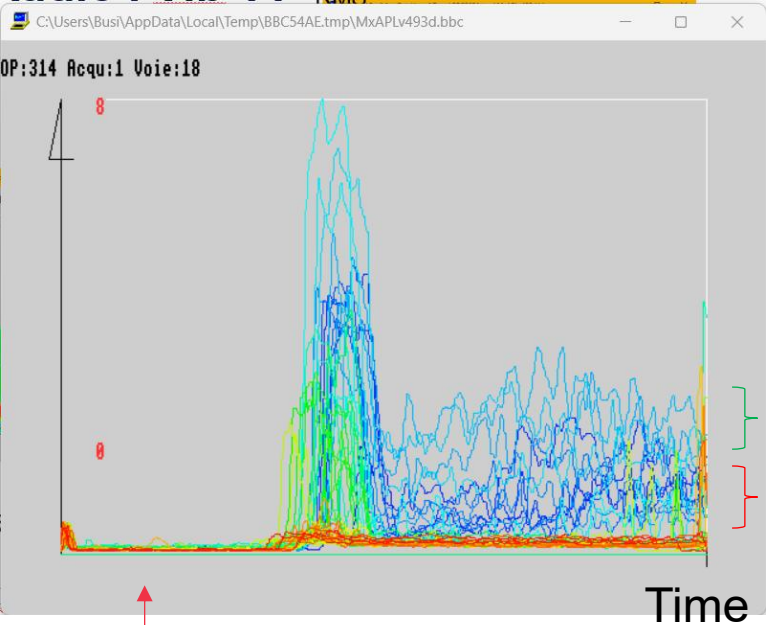
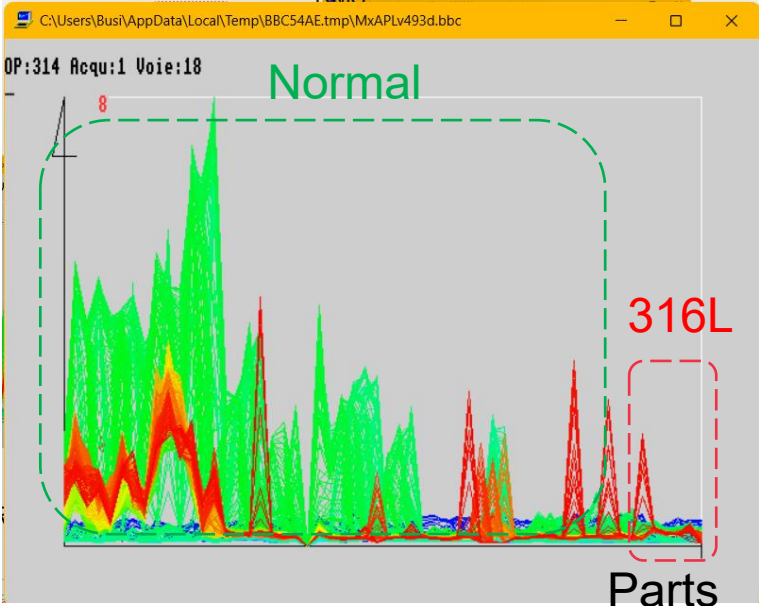


Warning to vertical offset : Spindle power increases when 316L grade
=> OK

Audio : Vib 11

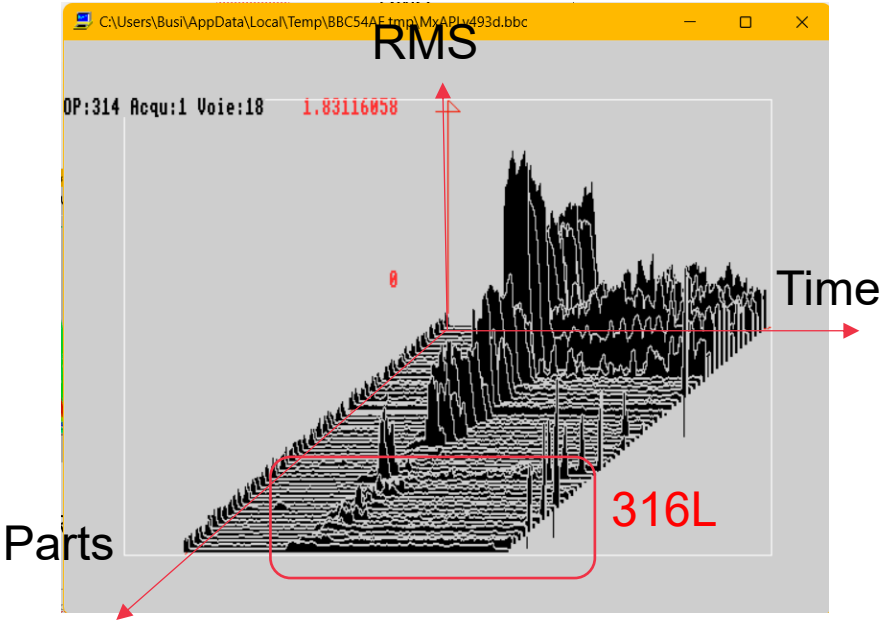
RMS

RMS



Normal

316L



Audio RMS, seems to decrease when 316L grade

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="250" description="AL F" monitor="true" tool="T02" Color="0;255;0" />
<Operation id="260" description="AL F" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="PERCAGE DIAM 10" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 315

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

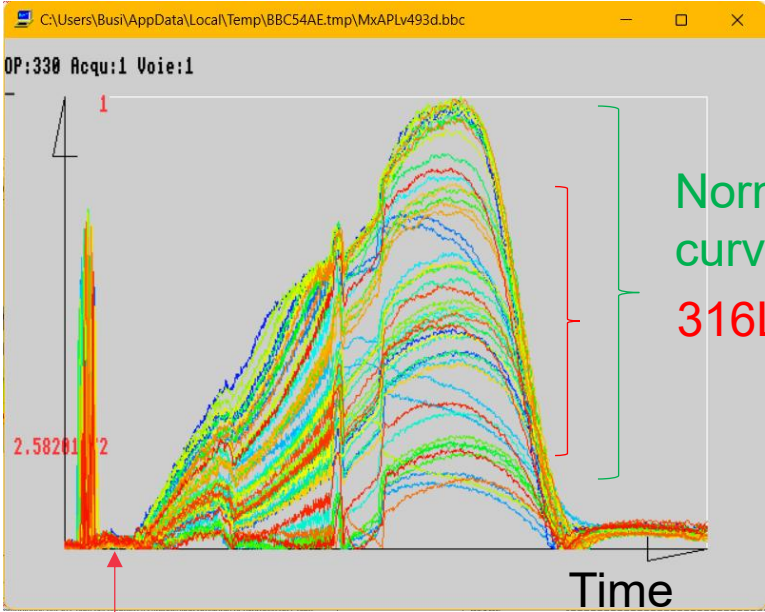
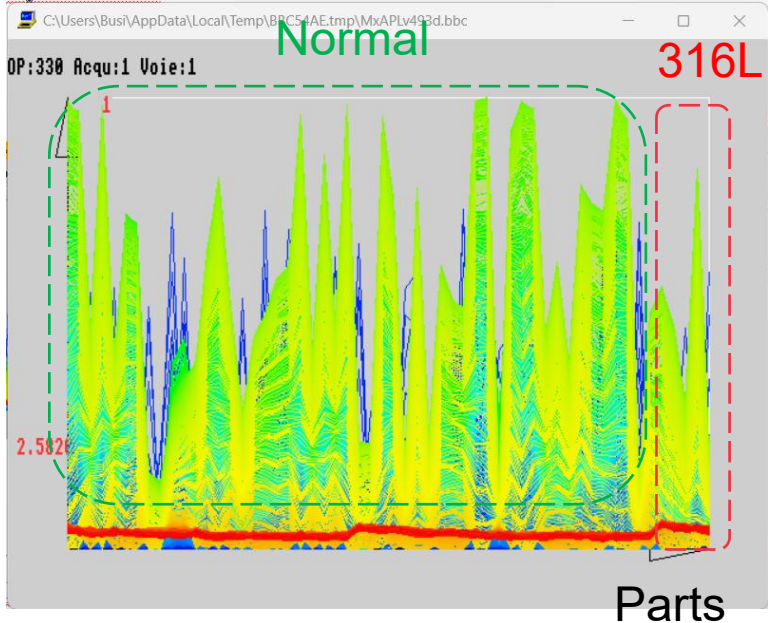
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Machining OP 330

Spindle Power : Ai1

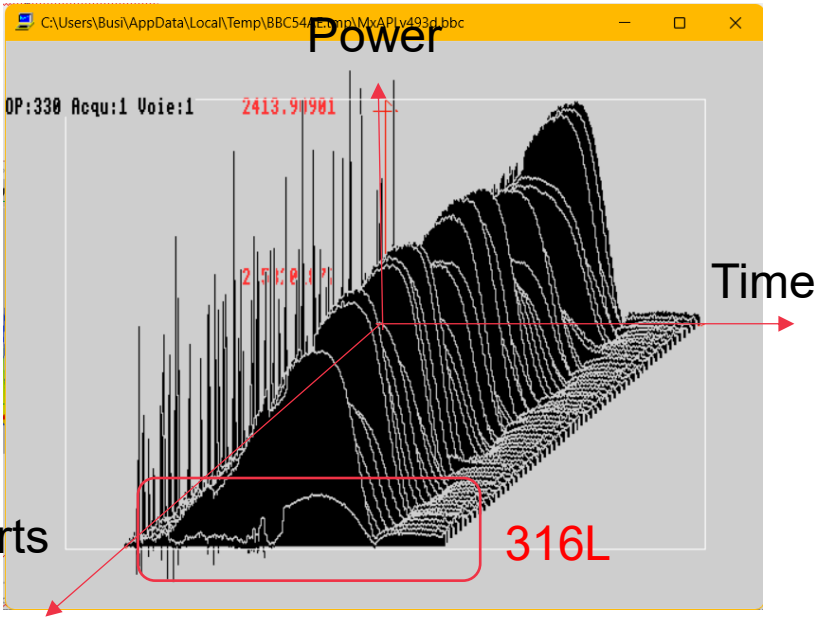
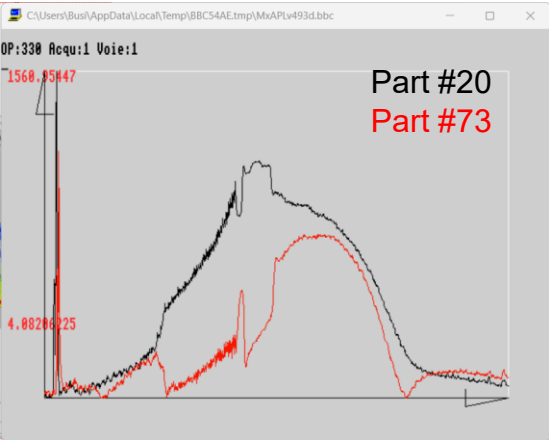
Power

Power



Before machining

In machining

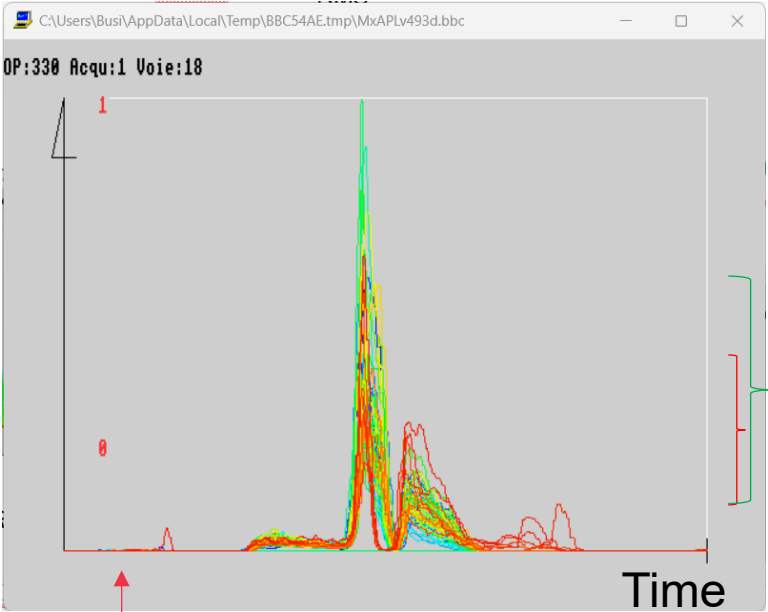
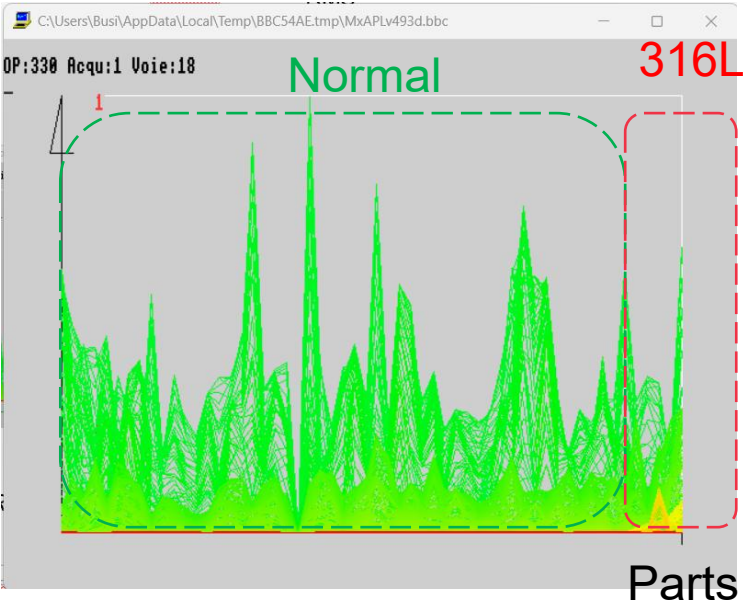


Spindle power remains the same when 316L grade ?

Audio : Vib 11

RMS

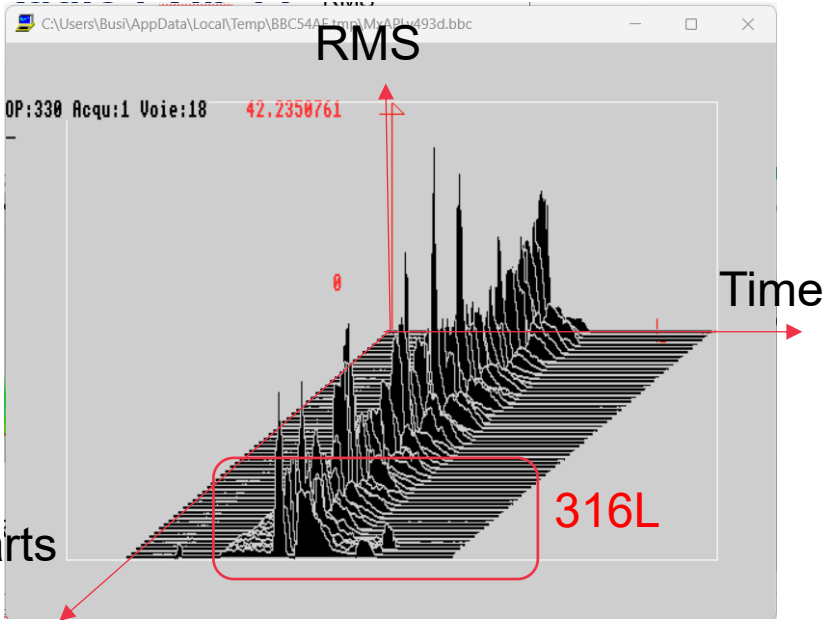
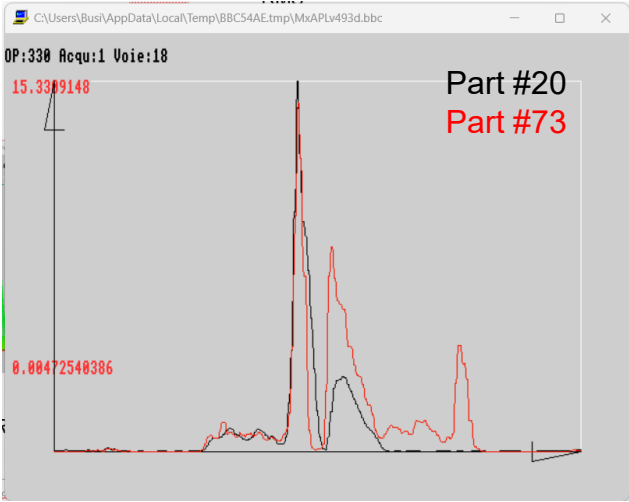
RMS



Before machining

In machining

Normal
316L



Audio RMS, remains the same when 316L grade



Pour un futur industriel
responsable et respectueux
de la planète